IAT New Test 25

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

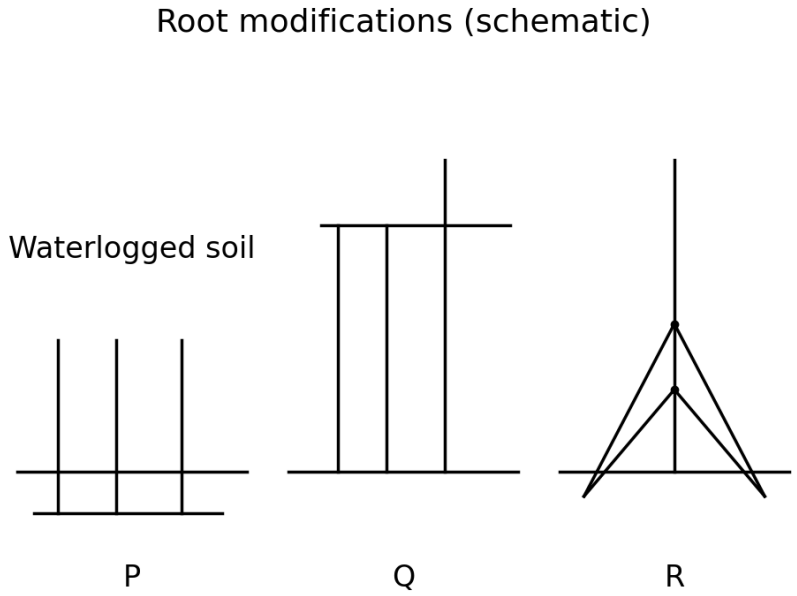
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A database lists Panthera leo and Panthera tigris under the same genus, while a lichen specimen consists of a fungal partner associated with a photosynthetic partner. Which conclusion follows without confusing naming rank with the number of biological partners?
A) The two Panthera names denote one species because the first word is shared
B) A lichen must be assigned to Panthera because it contains more than one cell type
C) The specific epithets alone establish that leo and tigris belong to different kingdoms
D) The two animals share a genus but are different named species; the lichen's association is not itself a taxonomic rank equivalent to genus

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Q2 The root-modification diagram shows P growing upward from waterlogged soil, Q descending from spreading branches into soil, and R arising from lower stem nodes obliquely into the soil. Which assignment of their principal functions is correct?



A) P mechanical support only; Q respiration only; R food storage
B) P gaseous exchange; Q prop-root support; R stilt-root support
C) P mineral-free photosynthesis; Q pollination; R seed dispersal
D) P absorption of atmospheric nitrogen gas; Q storage cotyledons; R tendrils

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Q3 A membrane protein has tags N and C. In native cells N faces the extracellular fluid and C faces the cytosol. Two preparations form sealed vesicles. External protease removes N but not C in P, and C but not N in Q. Detergent added before protease makes both tags disappear in both preparations. Protease cannot cross an intact bilayer and the tags are detected independently. Which inference uses all controls?
A) P retains native outside-out orientation and Q is inside-out; detergent results show that prior protection reflected inaccessibility rather than intrinsic protease resistance
B) P and Q both have native orientation; loss of different tags proves that the two proteins have different amino-acid sequences
C) Q must be freely permeable to protease, because one tag is lost; the protected tag can be ignored
D) Protease resistance of the protected tag is established permanently, so detergent must destroy the tag chemically without protease

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Q4 In an actively respiring mitochondrion, an enzyme converts succinate to fumarate and transfers electrons into the respiratory chain. Most other citric-acid-cycle enzymes are in the matrix. Which combination correctly identifies the exceptional enzyme location and distinguishes its role from substrate-level phosphorylation?
A) Succinate dehydrogenase is a soluble cytosolic enzyme that directly makes all mitochondrial ATP
B) Succinate dehydrogenase is in the outer membrane and directly synthesises GTP from succinate
C) Succinate dehydrogenase is associated with the inner mitochondrial membrane; substrate-level GTP formation occurs at the succinyl-CoA-to-succinate step instead
D) The citric acid cycle occurs entirely in the intermembrane space and contains no substrate-level phosphorylation

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Q5 After axons connecting the hypothalamus to the posterior pituitary are cut, posterior-pituitary hormone release is greatly reduced. Which statement correctly distinguishes the normal sites of synthesis and storage/release?
A) Oxytocin and vasopressin are normally synthesised by posterior-pituitary epithelial cells
B) These hormones are synthesised in hypothalamic neurons and transported to the posterior pituitary for storage and release
C) The anterior pituitary normally synthesises both hormones and sends them through the cut axons
D) The cut must stop thyroid follicular cells from synthesising thyroxine directly

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Q6 A primary oocyte is arrested in prophase I before birth. Near ovulation it completes meiosis I unequally, and the secondary oocyte arrests again until fertilisation. Which sequence correctly states the later arrest and the reason only one large functional female gamete is normally produced?
A) Metaphase I arrest; all cytoplasm is shared equally among four eggs
B) Prophase II arrest; DNA replication between meiotic divisions doubles egg number
C) Anaphase I arrest; polar bodies become equally large ova
D) Metaphase II arrest; unequal cytokinesis retains most cytoplasm in one cell, and meiosis II completes after sperm entry

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Q7 An RNA molecule in a laboratory reaction catalyses cleavage of another RNA, and its sequence is itself copied when the necessary machinery is supplied. Which inference is consistent with the RNA-world argument without exceeding this evidence?
A) RNA can combine informational and catalytic roles; the result does not imply that every modern organism has an RNA genome
B) RNA catalysis proves that DNA cannot store hereditary information
C) A catalytic RNA must be a protein because all biological catalysts contain amino-acid chains
D) The reaction establishes the complete historical sequence of the origin of the first cell

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Q8 A matched assay compares ingestion by phagocytes and blockade of microbial entry into target cells. Naive phagocytes ingest microbes A and B equally well. Serum from an A-immunised donor blocks entry of A but not B without phagocytes. Removing antibodies from that serum abolishes entry blockade; adding the isolated antibodies back restores it. Washed phagocytes from the immunised donor, tested without serum, ingest A and B at the same rates as naive phagocytes. Which conclusion and prediction are supported?
A) Prior immunisation has made the phagocytes specifically recognise only A; antibody removal should abolish all phagocytic ingestion
B) All observed protection is innate because a phagocyte can ingest either microbe; the serum-specific result is irrelevant
C) Broad phagocyte ingestion and A-specific antibody blockade are separable defences; removing antibodies need not remove ingestion, and these results do not establish antigen-specific memory in the washed phagocytes
D) The serum acts as an intact physical skin barrier; isolated antibodies cannot be responsible because no phagocytes are present

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Q9 DNA is to be isolated from bacterial, plant and fungal cells. After appropriate cell-wall treatment, all lysates contain DNA, RNA and protein. Which protocol matches the wall differences while preserving DNA?
A) Use cellulase for bacteria, lysozyme for fungi, then DNase to purify DNA
B) Use lysozyme for bacteria, cellulase for plant material and chitinase for fungi; remove RNA with RNase and protein with protease before precipitating DNA
C) Use chitinase for plants and RNase to remove all nucleic acids except protein
D) Add DNase before wall disruption because precipitation will reconstruct the hydrolysed DNA

Q10 A threatened wild plant is protected in its native forest, and seeds from several populations are also stored in a seed bank. Which interpretation correctly classifies these conservation actions?
A) Forest protection and seed banking are both in situ because both involve the same species
B) Seed banking alone preserves every ecological interaction of the native forest
C) Forest protection is ex situ because humans designate the protected area
D) Forest protection is in situ; seed banking is ex situ and preserves genetic material without recreating the entire native ecosystem

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Q11 A true-breeding red-flowered plant is crossed with a true-breeding white-flowered plant. F1 flowers are pink, and F1 selfing produces approximately one red:two pink:one white. Under a single-locus model, which conclusion separates allele segregation from the claim of complete dominance?
A) The alleles segregate normally, but the heterozygote has an intermediate phenotype rather than the phenotype of either homozygote
B) The 1:2:1 ratio proves that alleles blend and cannot separate again
C) Segregation is absent because the phenotype ratio is not 3:1
D) The F1 must contain only one allele because it has one visible colour

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Q12 Two skeletal muscle preparations have equal initial force. P has abundant myoglobin, mitochondria and capillaries; Q has much less of these. During prolonged repeated stimulation with adequate oxygen delivery, then during a brief interruption of oxygen supply, which comparison is most defensible?
A) Q must resist prolonged fatigue better because fewer mitochondria require less oxygen
B) P cannot use oxygen because myoglobin permanently traps it
C) P is better equipped for sustained oxidative activity; Q relies relatively more on anaerobic metabolism, and interruption of oxygen removes P's aerobic advantage without proving instant reversal of every performance measure
D) Both have identical fatigue behaviour because equal initial force determines endurance

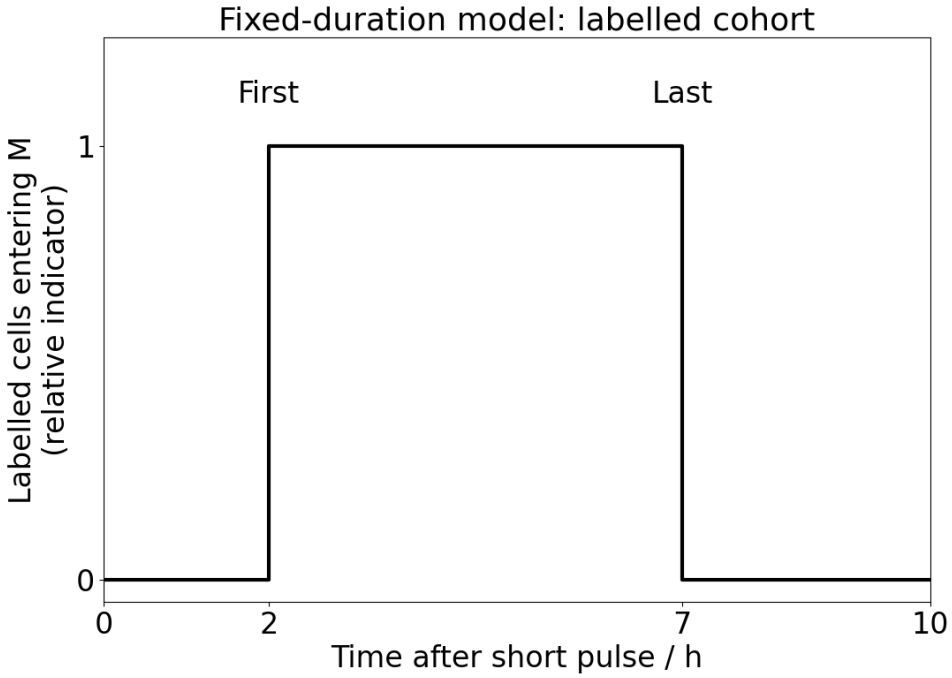
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Q13 In a dialysis model, blood and fresh dialysing fluid are separated by a membrane permeable to urea, water and small ions but impermeable to blood cells and large plasma proteins. The fluid has the same useful small-ion concentrations as blood but no urea. Pressure is balanced, and active transport is absent. Which initial net changes are expected?

A) Plasma proteins leave rapidly because their concentration is high, while urea stays in blood
B) Blood cells cross until cell numbers on both sides become equal
C) Every small ion necessarily leaves blood at the same net rate as urea
D) Urea diffuses into the fluid; useful ions have little initial net diffusion under the matched concentrations, and cells and large proteins remain in blood

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Q14 A very short DNA-label pulse marks cells currently in S phase. In a model culture every cell has the same fixed phase durations, all are cycling, and cells at every point of S phase are represented. The first labelled cells enter mitosis 2 h after the pulse and the last labelled cells enter mitosis 7 h after it. M lasts 1 h and the full cycle 12 h. Neglect pulse duration. What are S and G1 durations?



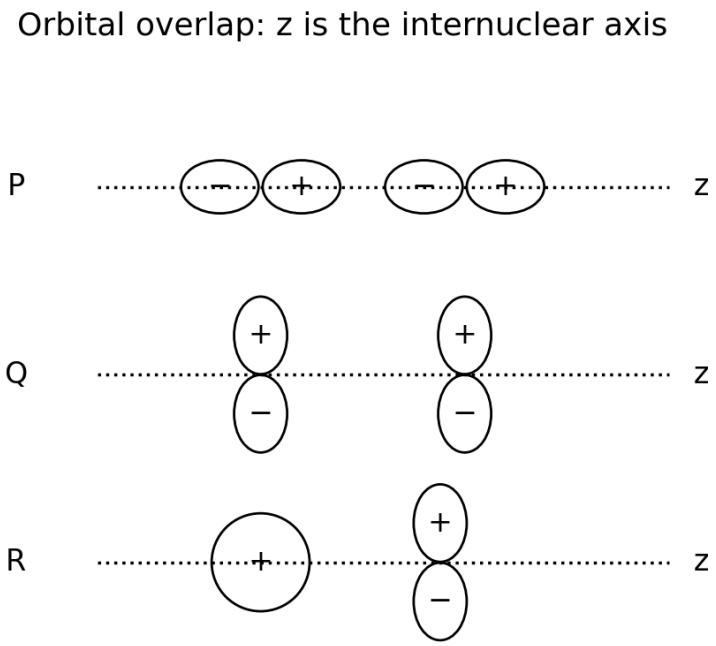
A) S=7 h; G1=2 h
B) S=5 h; G1=4 h
C) S=2 h; G1=7 h
D) S=5 h; G1=6 h

Q15 After delivery, a mother's mammary glands synthesise milk, but a temporary inhibition of oxytocin release prevents effective ejection during suckling. Which interpretation separates milk production from its delivery?

A) Prolactin and oxytocin are interchangeable names for the same hormone
B) Loss of oxytocin proves that no milk can have been synthesised
C) Prolactin supports milk production, while oxytocin normally promotes contraction of myoepithelial cells for milk ejection
D) Milk ejection is caused by inhibition of all mammary muscle contraction

Chemistry

Q16 The internuclear axis is z. The figure compares P: two pz orbitals approaching end-on; Q: two parallel px orbitals overlapping sideways; R: an s orbital facing a px orbital whose positive and negative lobes lie symmetrically on opposite sides of the z axis. Which idealised overlap assignment is correct?



A) P pi; Q sigma; R necessarily sigma
B) P and Q both sigma; R pi
C) P sigma; Q pi; R zero net overlap by symmetry
D) P zero; Q pi; R sigma

Q17 A colourless solvent contains only the two optical isomers of octahedral [Co(en)3]3+. At the chosen concentration, path length and wavelength, a pure solution of one isomer rotates +60 degrees and the other −60 degrees. Rotation is additive and the mixture gives +15 degrees. En is neutral and bidentate. What are the fraction of the positive isomer, cobalt oxidation state and metal coordination number?
A) 5/8; +3; 6
B) 3/8; +3; 3
C) 5/8; +2; 6
D) 3/4; +3; 6

Q18 For dissolution of one mole of an ionic solid, separating its gaseous ions costs +800 kJ mol−1 and hydration of those ions releases 760 kJ mol−1. At 298 K the dissolution entropy is +150 J mol−1 K−1. Use these data for the specified standard-state comparison. Which result is correct?
A) Dissolution enthalpy −40 kJ mol−1; Gibbs change −84.7 kJ mol−1
B) Dissolution enthalpy +40 kJ mol−1; Gibbs change −4.7 kJ mol−1
C) Dissolution enthalpy +1560 kJ mol−1; Gibbs change +1515.3 kJ mol−1
D) Dissolution enthalpy +40 kJ mol−1; Gibbs change +84.7 kJ mol−1

Q19 Ground-state hydrogen atoms are exposed separately to 12.0 eV photons and to electrons of kinetic energy 12.0 eV. The first excitation energy is 10.2 eV, the second is 12.09 eV and the ionisation energy is 13.6 eV. Ignore recoil, line broadening and all other processes. Which statement about a single absorption or inelastic collision is correct?
A) A 12.0 eV photon can supply 10.2 eV and retain 1.8 eV after being absorbed
B) Both probes must ionise hydrogen because 12.0 exceeds the first excitation energy
C) Neither can excite hydrogen because both energies must equal 13.6 eV
D) A photon cannot be absorbed into either listed excited level, but an electron can excite the first level and leave with 1.8 eV

Q20 A sodium-fusion extract from an organic compound may contain cyanide and sulfide as well as chloride. For the silver-nitrate halogen test, which preparation avoids a misleading precipitate?
A) Boil an appropriate portion with nitric acid before adding AgNO3; do not acidify it with HCl
B) Add HCl first because it removes chloride and cannot affect the test
C) Add AgNO3 directly because cyanide and sulfide cannot precipitate silver
D) Treat the original organic liquid with water only, since covalent chlorine is already a free chloride ion

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Q21 Benzoyl chloride is converted selectively into benzaldehyde using hydrogen and a deliberately deactivated palladium catalyst on BaSO4. Why is controlled catalyst activity important for this preparation?
A) It removes the benzene ring while leaving the acyl chloride unchanged
B) It allows reduction to stop at the aldehyde rather than promoting further reduction, so benzaldehyde retains an oxidisable CHO group
C) It converts the aldehyde into benzoic acid by supplying oxygen gas
D) It makes the product an amide by adding nitrogen from BaSO4

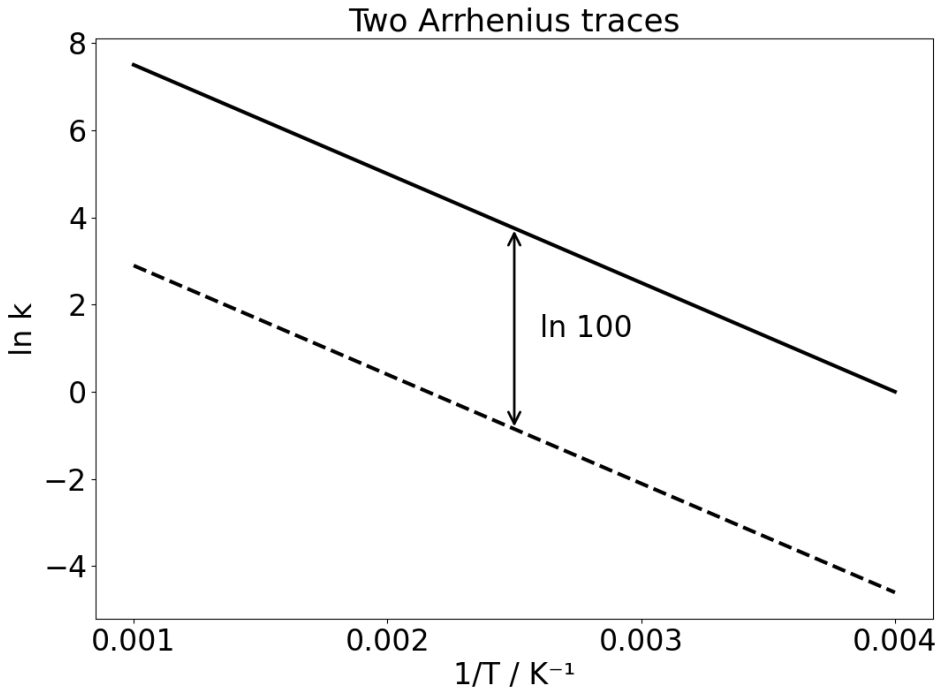
Q22 Benzenediazonium chloride whose two diazonium nitrogen atoms are both labelled 15N is divided into two portions. P undergoes azo coupling with phenoxide, followed by acidification; Q undergoes replacement with CuCl under the usual conditions. Assume the main para-coupled product in P and no isotope scrambling. Which fate of the labelled nitrogen is correct?
A) Both routes release the two labels as nitrogen gas, leaving no nitrogen in either organic product
B) P retains only one labelled nitrogen in an amine; Q retains both in a nitrile
C) P retains both labels in the azo linkage of mainly p-hydroxyazobenzene; Q releases them as labelled N2 and gives chlorobenzene
D) P gives chlorobenzene with both labels attached to chlorine; Q gives the azo dye

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Q23 A hydrolysate contains nucleotides from only double-stranded DNA and single-stranded RNA. Per 100 nucleotide units, the pure DNA component has A20,T20,G30,C30, and the pure RNA component A25,U15,G35,C25. The mixture contains T8 per 100 units. What fraction of units came from DNA, and is total G necessarily equal to total C?
A) 0.40; no, the mixture has G33 and C27 per 100 units
B) 0.40; yes, every nucleic-acid mixture obeys G=C
C) 0.60; no, the mixture has G32 and C28 per 100 units
D) 0.80; yes, because thymine and uracil must be equally abundant

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Q24 Two reactions have parallel straight plots of ln k against 1/T over the same range. The upper line is higher by ln 100 at every temperature. Using the Arrhenius model, which comparison is correct?



A) The upper reaction has activation energy 100 times larger and the same prefactor
B) The upper reaction has a lower activation energy because its line is vertically higher
C) Both prefactors are equal, so the two rate constants must be equal
D) The activation energies are equal, and the upper reaction has a prefactor 100 times larger

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Q25 A closed vessel at chemical equilibrium contains nitrogen, hydrogen and ammonia. A tracer amount of 15N2 is introduced without appreciably changing temperature, volume or concentrations. Later some labelled ammonia is detected while the macroscopic concentrations remain effectively constant. Which inference is justified?
A) The forward reaction occurs but the reverse rate must be exactly zero
B) Microscopic forward and reverse reactions continue at equal average rates; constant bulk composition does not mean individual molecules never react
C) The catalyst, if present, must have changed the equilibrium constant to create the label
D) Detection of label proves that atom conservation fails at equilibrium

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Q26 In a sealed vessel without electrodes, 0.100 mol H2O2 decomposes completely by disproportionation: 2H2O2 gives 2H2O and O2. How many moles of O2 form, and how many moles of electrons are transferred internally from the oxidised oxygen atoms to the reduced ones?
A) 0.100 mol O2; 0.200 mol electrons
B) 0.0500 mol O2; zero electron transfer because there is no external current
C) 0.0250 mol O2; 0.100 mol electrons
D) 0.0500 mol O2; 0.100 mol electrons, without requiring an external circuit current

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Q27 One mole of a reaction proceeds isothermally at 400 K in contact with a 400 K reservoir. Its enthalpy change is −48 kJ mol−1 and system entropy change is −100 J mol−1 K−1. Only pressure–volume work occurs. Treat these changes as temperature independent for a second comparison at 600 K. Which statement is correct?
A) At 400 K total entropy decreases by 20 J K−1, so the reaction is unfavourable
B) At 400 K total entropy increases by 220 J K−1; at 600 K it must also increase
C) At 400 K total entropy increases by 20 J K−1 and delta G=−8 kJ mol−1; at 600 K the same reaction has delta G=+12 kJ mol−1
D) At both temperatures spontaneity follows from negative enthalpy alone

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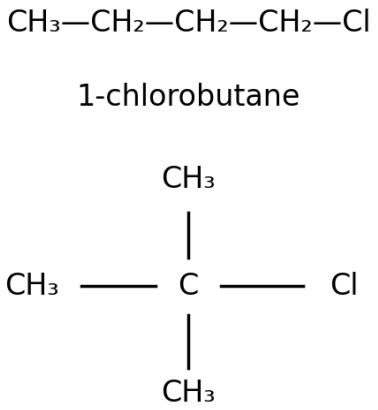
Q28 At the same temperature, 100 mL of a 20.0% by-mass solution of neutral solute X has density 1.20 g mL−1; 100 mL of a 10.0% solution has density 1.00 g mL−1. They are mixed without reaction or loss. The measured final density is 1.25 g mL−1. M(X)=100 g mol−1. What are the final molarity and solute mass percentage? Do not assume additive volumes.
A) 1.70 M; 15.0%
B) 85/44 M; 170/11% (approximately 1.932 M; 15.45%)
C) 85/44 M; 30.0%
D) 1.70 M; 170/11%

Q29 Benzene and cyclohexene are tested separately. Compare complete catalytic hydrogenation to cyclohexane and a room-temperature bromine test without a Lewis-acid catalyst or light. Which pair of observations is expected?
A) Benzene consumes 3 mol H2 per mole but does not readily decolourise bromine in this test; cyclohexene consumes 1 mol H2 and decolourises bromine by addition
B) Both consume 3 mol H2 and both give immediate bromine addition under these conditions
C) Benzene consumes 1 mol H2 and cyclohexene 3 mol H2
D) Neither can react with hydrogen because cyclic compounds contain no unsaturation

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Q30 The diagram shows 1-chlorobutane and 2-chloro-2-methylpropane, which have the same molecular formula. At the same external pressure, the branched isomer has a lower boiling point. In a separate polar protic solvolysis it reacts more readily by an SN1 pathway. Which explanation correctly separates these observations?

Isomeric haloalkane connectivity



A) The lower boiling point proves that the tertiary carbocation is less stable
B) Faster SN1 reaction requires a higher boiling point because both depend on the same transition state
C) The branched molecule has lower molar mass, and that alone explains both observations
D) Branching reduces effective intermolecular contact, while tertiary carbocation stability favours SN1; a physical-property ranking need not match a reaction-rate ranking

Mathematics

Q31 A nondegenerate conic is defined by distance to F=(1,0) equal to e times perpendicular distance to x=2. It passes through the origin and has e>0. What are its eccentricity and centre?
A) e=2; centre (4/3,0)
B) e=1/2; centre (2/3,0)
C) e=1/2; centre (1,0)
D) e=1; no finite centre

Q32 At a point (x,1/x) on y=1/x with x>0, the tangent and normal meet the x-axis at equal distances from the origin. What is x^4?
A) 3
B) 1
C) 1/2
D) 1/3

Q33 A rectangle has perimeter 12 and is required to have its length strictly greater than its width, with both dimensions positive. What is the supremum of its area, and is it attained?
A) 6; attained by a 4 by 2 rectangle
B) 9; attained by the square
C) 9; not attained under the strict length–width requirement
D) 12; not attained because the width tends to zero

Q34 On [0,2], define f(x)=(x^3−x^2−x+1)/[(x−1)(x^2+1)] for x not equal to 1, and define f(1)=7. What is the integral from 0 to 2 of |f(x)| dx?
A) pi−2 arctan 2
B) 2−2 arctan 2
C) 2 arctan 2
D) 7+pi−2 arctan 2

Q35 On S={1,2,3,4}, let f send 1 to 2, 2 to 3, 3 to 1 and 4 to 4. How many maps g:S to S satisfy g composed with f = f composed with g, and how many of these are bijections?
A) 3 maps; 3 bijections
B) 4 maps; 4 bijections
C) 4 maps; 3 bijections
D) 16 maps; 6 bijections

Q36 Let e1,e2,e3 be a right-handed orthonormal basis with e3=e1 cross e2. A vector v satisfies v+e1 cross v=2e2. What is v?
A) e2+e3
B) e2−e3
C) 2e2−2e3
D) e1+e2−e3

Q37 Let A have rows (2,1),(0,2). For a positive integer n, the upper-right entry of A^n is three times either diagonal entry. What are n and trace(A^n)?
A) n=3; trace 16
B) n=6; trace 64
C) n=3; trace 128
D) n=6; trace 128

Q38 The numbers 1,2,3,4,5 are randomly ordered and revealed one at a time. Revealing stops as soon as the running sum reaches or exceeds 6. Given that exactly two numbers were revealed, what is the probability that the first number was 5?
A) 1/3
B) 1/5
C) 1/2
D) 1/4

Q39 Four people are chosen from six married couples. How many committees contain exactly one complete couple, with the other two members drawn from two different remaining couples?
A) 480
B) 120
C) 240
D) 360

Q40 Let P(x)=(1+x)^5(1−x+x²−x³+...+x^10). What is the coefficient of x^12 minus the coefficient of x^10 in P(x)?
A) −4
B) 0
C) 5
D) 4

Q41 Three nonzero real numbers a,b,c are consecutive terms of a geometric progression. Their positive geometric mean sqrt(ac) is 6, their sum a+b+c is 9, and a+c=15. The common ratio has magnitude greater than 1. What are the ratio and the sum of the first four terms of this progression?
A) r=−2; sum −15
B) r=2; sum 45
C) r=−1/2; sum 15/2
D) r=−2; sum 33

Q42 For real a, x²−2ax+(a+3)=0 has two nonreal conjugate roots, both of modulus 2. What are a and the principal argument of the root with positive imaginary part?
A) a=−1; argument 2pi/3
B) a=1; argument 2pi/3
C) a=1; argument pi/3
D) a=3; argument pi/6

Q43 Five real observations lie in [−2,3] and have mean zero. What is the greatest possible mean deviation about zero?
A) 2
B) 12/5
C) 3
D) 5/2

Q44 For x>1, solve y’=y/(x ln x) with y(e)=2. Which statement gives the solution on the maximal interval containing e within the differential equation’s domain?
A) y=2 ln x on (1,infinity); its finite limit zero at 1 does not make the differential equation defined there
B) y=2x on (0,infinity)
C) y=2 ln x on (0,infinity), with the same equation valid at x=1
D) y=2/ln x on (1,infinity)

Q45 Let f:[0,infinity) to [0,infinity) be x^2 for $0 \le x \le 1$ and $2x-1$ for $x>1$. Which statement about its inverse g is correct?

A) g is discontinuous at 1 because its two formulas differ

B) g has finite right derivative 0 at zero and derivative 1 at 1

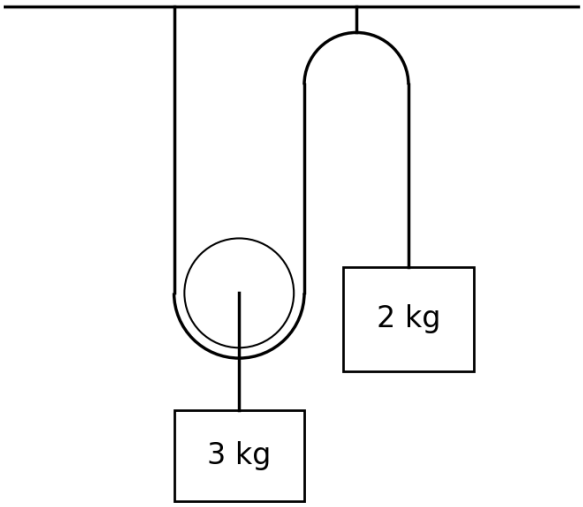
C) g does not exist because x^2 is not one-one on all real numbers

D) g is continuous on its domain, has derivative 1/2 at 1, and has no finite right derivative at zero

Physics

Q46 A massless movable pulley supports a 3.0 kg load. One end of its light rope is fixed to the ceiling; the other passes over a fixed pulley and carries a 2.0 kg mass. Pulleys are frictionless and rope does not slip. Take g=10 m s−2. What are the accelerations of the 3.0 kg load and 2.0 kg mass?

One movable pulley and one fixed pulley

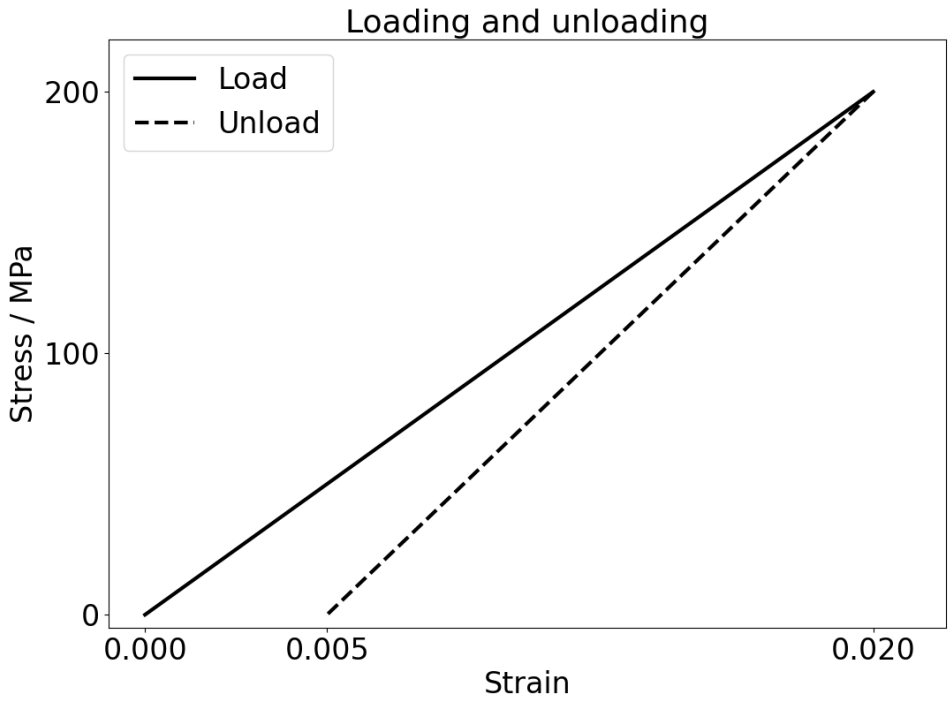


A) 10/11 m s−2 upward; 20/11 m s−2 downward
B) 2 m s−2 downward; 2 m s−2 upward
C) 10/7 m s−2 upward; 10/7 m s−2 downward
D) Zero for both because 3 kg exceeds 2 kg

Q47 A 2.0 kg vehicle moves in a straight line at positive speed v. Its engine power is $P(v)=v^2+6v$ watts when v is in m s−1, and the opposing drag is v newtons. It starts at 2.0 m s−1. What is its speed after travelling 10 m, and is its acceleration constant in this model?
A) 6.0 m s−1; no
B) 8.0 m s−1; yes, 3.0 m s−2
C) 8.0 m s−1; no, constant power means changing acceleration
D) sqrt(34) m s−1; yes, 1.5 m s−2

Q48 Two thin concentric spherical shells have radii R and 3R and unknown positive masses m1,m2. Potential is zero at infinity. Measurements give potential −5GM/R at the centre and −3GM/R at r=2R, where M is a reference mass. What are m1,m2, and the quasistatic external work in moving a test mass between any two points strictly inside the inner shell?
A) m1=2M,m2=9M; zero
B) m1=4M,m2=3M; nonzero because potential is negative
C) m1=4M,m2=3M; zero
D) m1=5M,m2=0; zero

Q49 A specimen loads along a straight stress–strain line from (0,0) to strain 0.020 and stress 200 MPa. It unloads along a straight line to zero stress at strain 0.005, as shown. Its initial volume is $2.0\times10-6\ m^3$. Neglect volume change and assume all unrecovered work is dissipated, with no other stored-energy change. How much energy is dissipated per cycle, and what residual strain remains?



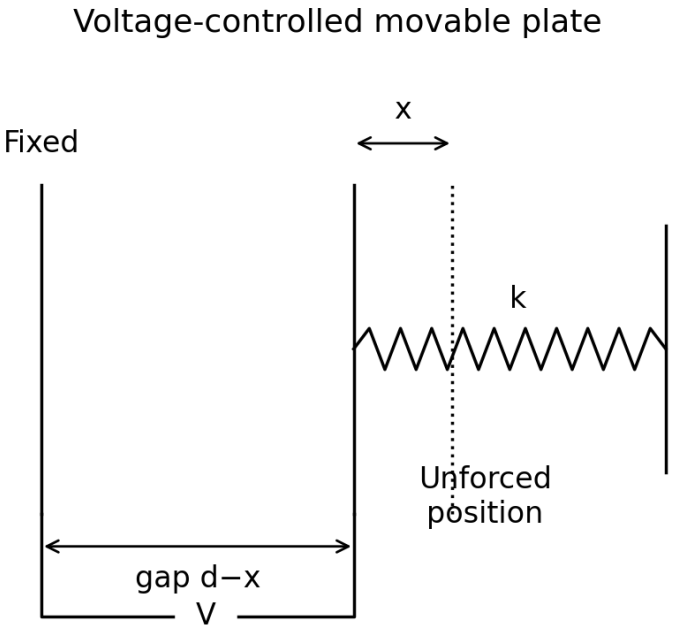
A) 4.0 J; zero
B) 3.0 J; 0.005
C) 1.0 J; zero
D) 1.0 J; 0.005

Q50 An expansion-compensating frame produces a measured separation X=L1−L2. The two rods are arranged to contribute oppositely, remain at the same temperature, and expand freely to first order. Their coefficients are alpha1=1.0×10−5 K−1 and alpha2=2.0×10−5 K−1. If X is to equal 1.0 m and be temperature independent to first order, what initial lengths are required?
A) L1=1.0 m,L2=2.0 m
B) L1=2.0 m,L2=1.0 m
C) L1=1.5 m,L2=0.5 m
D) L1=L2=1.0 m

Q51 In a dilute gas of identical hard-sphere molecules in a fixed volume, number density doubles while temperature falls to one-quarter. Molecular mass and collision cross-section are unchanged. In the kinetic-theory model, how do collision frequency per molecule and total collision events per unit volume per time change?
A) Both stay unchanged
B) Both double
C) Per molecule halves; per volume stays unchanged
D) Per molecule stays unchanged; per volume doubles

Q52 A 0.20 kg particle executes one-dimensional SHM. At one instant its displacement is 0.030 m and speed 0.40 m s−1; at another its displacement is 0.040 m and speed 0.30 m s−1. What is its total mechanical energy?
A) 0.025 J
B) 0.050 J
C) 0.016 J
D) 0.009 J

Q53 A parallel-plate capacitor has fixed plate area A. One plate is supported by a spring of constant k; the unforced gap is d and inward displacement is x. An ideal battery holds voltage V. Ignore fringing and use electric attraction F=epsilon0 A V²/[2(d−x)²]. What is the stable-equilibrium condition and limiting pull-in voltage squared?

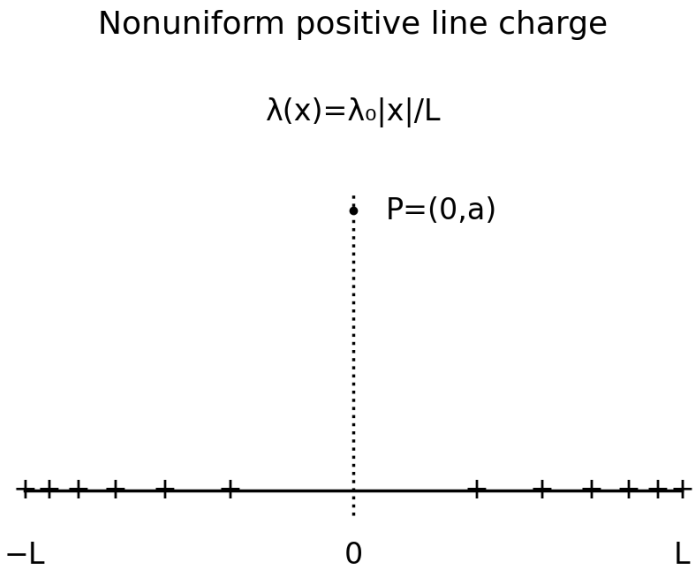


A) Stable for every 0<x<d; Vlimit²=2kd³/(epsilon0 A)
B) Stable for x>d/3; Vlimit²=8kd³/(27 epsilon0 A)
C) Stable for 0<x<d/3; Vlimit²=8kd³/(27 epsilon0 A)
D) Stable for 0<x<d/2; Vlimit²=kd³/(4 epsilon0 A)

Q54 Three samples have measured small-field responses: P has weak negative susceptibility with little temperature variation; Q has small positive susceptibility approximately proportional to 1/T; R has strong positive response and remanent magnetisation below a characteristic temperature, losing remanence above it. Which classification and permeability comparison is correct?
A) P diamagnetic, Q paramagnetic, R ferromagnetic below its transition; P has relative permeability below 1
B) P paramagnetic, Q diamagnetic, R an intrinsic semiconductor
C) P ferromagnetic, Q paramagnetic, R diamagnetic; all relative permeabilities are below 1
D) P and Q must be identical because both have small susceptibility

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Q55 A positively charged thin rod lies on the x-axis from −L to L. Its line charge density is lambda(x)=lambda0|x|/L with lambda0>0. The point P is (0,a), a>0, as shown. Which direction does the rod's electric field at P have?



A) Along +y; paired horizontal components cancel and paired vertical components add
B) Along −y because the density vanishes at the rod's centre
C) Zero because every source element has a symmetric partner
D) Along +x because the density rises toward the right end

Q56 A cell has constant emf 6.0 V and internal resistance 1.0 ohm. A fixed external resistor must receive 40 J in 10 s. Two positive resistances can meet the requirement. Which resistance minimises the total energy supplied by the cell, and what is that source energy?
A) R=(7−3sqrt(5))/2 ohm; source energy 40 J
B) R=1 ohm; source energy 80 J
C) R=(7+3sqrt(5))/2 ohm; source energy 180−60sqrt(5) J
D) R=(7+3sqrt(5))/2 ohm; source energy 40 J

Q57 A thin convex lens of focal length 5.0 cm is used as a simple magnifier. The eye is 5.0 cm behind the lens and its near point is 25 cm from the eye. The final virtual image is at that near point on the object side. What object distance from the lens and angular magnification relative to unaided viewing at 25 cm are obtained? Use small angles.
A) 25/6 cm; magnification 6
B) 4.0 cm; magnification 5
C) 4.0 cm; magnification 6
D) 5.0 cm; magnification 5

Q58 A metal has work function 2.0 eV. A voltmeter reports stopping potential as the true value plus a fixed +0.40 V contact offset. Use hc=1240 eV nm. For 400 nm illumination, what reading is reported, and which of 800 nm infrared or 300 nm ultraviolet can cause emission in this ideal one-photon model?
A) 1.10 V; both wavelengths
B) 1.50 V; both wavelengths
C) 0.70 V; only infrared
D) 1.50 V; only ultraviolet

Q59 Hydrogen atoms are initially in n=2. Each absorbs one resonant 1.89 eV photon to reach n=3, then eventually returns to n=1 by radiative emission. The excitation energies above n=1 are 10.20 eV for n=2 and 12.09 eV for n=3. What total energy can one such atom emit during its return from n=3 to n=1, and why can it exceed the newly absorbed energy?
A) 1.89 eV only, because no atom can emit more than the most recently absorbed photon
B) 12.09 eV, because 10.20 eV was already stored in its initial excitation
C) 10.20 eV, because the new photon energy must be discarded
D) 24.18 eV, because two-step decay always doubles the level gap

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Q60 An ideal AC generator rotates at fixed angular speed in a uniform magnetic field and feeds a pure resistor. After the coil turns through half a revolution, its induced emf and current reverse. What happens to the electromagnetic torque and mechanical power needed from the drive, away from the zero-emf instants?
A) Torque reverses with current, so the resistor alternately drives and brakes the rotor
B) Torque vanishes whenever current becomes negative
C) The torque continues to oppose rotation and the drive power remains positive because it supplies Joule heat
D) The drive power becomes negative for half of every cycle although I^2R remains positive

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / D Shared binomial genus and a lichen association describe different classification and ecological relations

Answer: (D) A shared generic name and a symbiotic association describe different kinds of organisation.

Why: In binomial nomenclature the first word identifies the genus and the second contributes the specific epithet. The two listed animals therefore share Panthera but have different complete species names. A lichen is an association of fungal and photosynthetic partners, not a new rank in the genus–species hierarchy. Its composite organisation cannot be inferred from, or used to redefine, the naming rules for the animals.

Why not the others: (A) treats sharing a genus as identity of species. (B) invents a relation between a lichen association and an animal genus. (C) tries to infer a kingdom distinction from specific epithets despite the common genus.

Key point: Keep the rank encoded by a scientific name separate from the ecological relationship among organisms.

Q2 / B Respiratory, prop and stilt roots distinguished by origin and environmental function

Answer: (B) The upward respiratory roots differ from two mechanically supportive root arrangements.

Why: Waterlogged soil restricts gaseous exchange, and pneumatophores projecting above it provide an aerating route. Prop roots descend from branches and can support a broad crown, as in banyan. Stilt roots arise from lower stem nodes and enter the soil obliquely, as in maize or sugarcane. Their position of origin and direction are therefore relevant to function; a root projecting upward is not simply another support column.

Why not the others: (A) exchanges the gas-exchange role of P with Q's supporting geometry. (C) attributes reproductive or photosynthetic functions not supported by these root forms. (D) confuses gas exchange with direct nitrogen-gas assimilation and labels roots as cotyledons or tendrils.

Key point: Infer root modification from its origin, orientation and environmental constraint together.

Q3 / A Protease access and detergent rescue distinguish opposite vesicle orientations from protein resistance or membrane leakage

Answer: (A) The two sealed preparations expose opposite native faces, and detergent removes the access barrier.

Why: P exposes N, the region facing extracellular fluid in the native cell, while protecting the cytosolic C tag. That is the native outside-out arrangement. Q instead exposes C and protects N, which fits an inside-out orientation rather than complete membrane leakage. In each preparation the previously protected tag becomes protease-sensitive after detergent disrupts the bilayer. That control separates a physical accessibility difference from an intrinsically uncleavable tag. Independent tag detection prevents disappearance of one signal from being mistaken for loss of the entire protein.

Why not the others: (B) ignores the known native topology and invents sequence differences that are unnecessary to explain the opposite access. (C) predicts access to both tags if the membrane were freely permeable, contrary to the retained protected tag. (D) contradicts the detergent-plus-protease result by treating a reversible access restriction as permanent chemical resistance.

Key point: Combine sidedness, impermeability and a barrier-removal control before inferring

membrane-protein orientation.

Q4 / C Membrane-associated succinate dehydrogenase distinguished from the separate substrate-level GTP step

Answer: (C) Succinate oxidation is membrane linked, whereas a different cycle step yields substrate-level GTP.

Why: Succinate dehydrogenase participates in the citric acid cycle and in electron transfer associated with the inner mitochondrial membrane. It is the location exception to the mainly matrix-localised cycle enzymes. The conversion of succinyl-CoA to succinate instead couples a high-energy substrate transformation to nucleotide phosphorylation. Electron delivery to the respiratory chain and direct substrate-level phosphorylation are distinct contributions to respiration, even though both occur within the overall mitochondrial pathway.

Why not the others: (A) gives the enzyme the wrong compartment and treats one step as the source of all ATP. (B) places it in the wrong membrane and confuses it with the GTP-forming step. (D) relocates the whole cycle and denies a retained phosphorylation reaction.

Key point: Locate individual respiratory reactions and distinguish electron transfer from direct nucleotide phosphorylation.

Q5 / B Hypothalamic synthesis distinguished from posterior-pituitary storage and release

Answer: (B) Hypothalamic synthesis is separated from posterior-pituitary storage and release.

Why: Oxytocin and vasopressin are produced by hypothalamic neurosecretory neurons. Their axons carry the hormones to endings associated with the posterior pituitary, where they are stored and released into the circulation. Cutting this connection therefore disrupts the delivery/release arrangement without making the posterior pituitary the original synthesis site. The result concerns this neural pathway rather than every endocrine gland.

Why not the others: (A) assigns synthesis to the storage/release region instead of hypothalamic neurons. (C) substitutes the anterior pituitary for the neuronal source. (D) claims a direct block in a separate gland without evidence connecting it to the severed axons.

Key point: The organ releasing a hormone need not be the tissue that synthesised it.

Q6 / D Oocyte arrest timing and unequal cytoplasmic allocation explain one large functional gamete

Answer: (D) The ovulated secondary oocyte is normally arrested at metaphase II and retains most cytoplasm.

Why: Completion of meiosis I separates homologous chromosomes while dividing the cytoplasm unequally. The secondary oocyte keeps most of that cytoplasm, whereas a small polar body receives the other chromosome set. It enters meiosis II and arrests at metaphase II; sperm entry triggers completion. This arrangement preserves cellular resources in one prospective ovum rather than distributing them among four equally large functional eggs.

Why not the others: (A) names the wrong arrest and replaces unequal cytokinesis with equal division. (B) invents replication between the two meiotic divisions and gives the wrong phase. (C) places the arrest during chromosome separation and wrongly promotes polar bodies into equivalent ova.

Key point: Follow meiotic stage and cytoplasmic allocation separately in oogenesis.

Q7 / A RNA information and catalysis support an RNA-world possibility without a universal modern-genome claim

Answer: (A) The demonstrated dual capabilities support a possible early role for RNA without establishing a universal modern genome or complete history.

Why: An RNA sequence can encode information, and some RNA molecules also catalyse reactions. Combining these properties makes RNA a plausible participant in early systems before the modern division of labour among DNA, RNA and proteins. The experiment demonstrates capabilities under supplied conditions, not the genome type of every present-day organism. It also does not reconstruct all steps by which autonomous living systems originated.

Why not the others: (B) denies DNA's established informational role without a relevant test. (C) mistakes the chemical nature of a ribozyme for a polypeptide enzyme. (D) turns a limited molecular observation into proof of an entire historical pathway.

Key point: Distinguish evidence for a molecule's possible evolutionary role from proof of a complete origin scenario.

Q8 / C Antibody removal and add-back distinguish specific serum blockade from unchanged broad phagocyte ingestion

Answer: (C) The data separate a broad cellular innate function from a specific antibody-mediated acquired function.

Why: Naive and donor phagocytes show the same ingestion of A and B, so these measurements do not demonstrate an A-specific change in the washed phagocytes. By contrast, the donor serum blocks only A, and loss after antibody removal followed by rescue on add-back ties that measured blockade to antibodies. The two assays measure different processes, ingestion and entry inhibition; removing one mechanism therefore need not abolish the other. Prior immunisation is relevant to the antibody specificity, but it does not justify assigning every donor-cell effect to memory.

Why not the others: (A) contradicts equal A/B ingestion and extends antibody depletion to an independently measured cellular function. (B) ignores the antigen-specific serum effect and its removal/rescue controls. (D) confuses a soluble molecular defence with skin and denies the demonstrated antibody rescue.

Key point: Use selective removal and rescue in matched assays to distinguish coexisting innate and acquired mechanisms.

Q9 / B Wall-specific lysis and selective nuclease cleanup distinguish DNA isolation from target destruction

Answer: (B) Wall-specific enzymes are followed by selective removal of contaminating RNA and protein.

Why: Bacterial, plant and fungal walls have different major structural materials, so lysozyme, cellulase and chitinase are appropriate respective choices in the standard isolation context. Once cells are lysed, RNase degrades RNA and protease removes proteins while DNA is retained. Chilled ethanol can then precipitate DNA from the prepared material. A precipitation step separates molecules; it cannot reverse nuclease hydrolysis of the desired DNA.

Why not the others: (A) mismatches walls and destroys the target with DNase. (C) assigns a fungal-wall enzyme to plant cellulose and confuses RNA removal with elimination of every nucleic acid. (D) intentionally hydrolyses the target and assumes precipitation repairs it.

Key point: Match the lysis enzyme to the wall and the cleanup enzyme to the contaminant, preserving the molecule sought.

Q10 / D Native-population protection and seed banking conserve different levels of biological context

Answer: (D) Protecting the natural population differs from storing resources outside its habitat.

Why: In situ conservation maintains organisms within their natural ecological setting, where interactions and environmental processes can continue. A seed bank holds material outside that setting and is therefore an ex situ measure. Sampling several populations can retain more genetic representation, but storage does not reproduce pollination, competition and other forest interactions. The two approaches can complement one another without being ecologically identical.

Why not the others: (A) classifies by species identity instead of conservation setting. (B) attributes a complete ecosystem to stored seeds. (C) confuses human management with the location of conservation relative to the native habitat.

Key point: Distinguish where conservation occurs from the identity of the species being conserved.

Q11 / A Incomplete dominance alters phenotype classes without invalidating Mendelian allele segregation

Answer: (A) Incomplete dominance changes genotype-to-phenotype mapping without abolishing segregation.

Why: The F1 heterozygote transmits either parental allele, and selfing yields the familiar homozygote:heterozygote:homozygote genotype ratio of 1:2:1. When the heterozygote is phenotypically distinct, that genotype ratio is visible directly as three flower-colour classes. A 3:1 phenotype ratio requires one allele to mask the other completely, which is not the observation here. Recovery of both parental colours in F2 is inconsistent with irreversible blending.

Why not the others: (B) ignores the reappearance of the parental phenotypes. (C) treats complete dominance as a requirement for segregation. (D) confuses one observed colour with one copy of a gene in a diploid cell.

Key point: Test allele transmission separately from how heterozygotes express the transmitted alleles.

Q12 / C Red and white fibre structure predicts oxygen-dependent endurance rather than an unconditional force ranking

Answer: (C) The structural evidence predicts different metabolic capacities rather than a universal force ranking.

Why: Myoglobin supports oxygen availability within muscle, and abundant mitochondria and capillaries favour sustained aerobic ATP production in red fibres. White fibres rely relatively more on anaerobic routes and tend to fatigue sooner during prolonged activity under suitable oxygen supply. Removing that supply limits the oxidative route, so P's advantage cannot simply be assumed to persist unchanged. Equal initial force does not establish equal energy reserves, metabolic capacity or later endurance.

Why not the others: (A) treats limited oxidative machinery as an endurance advantage under adequate oxygen. (B) denies reversible oxygen binding and release by myoglobin. (D) infers the entire fatigue time course from one initial mechanical measurement.

Key point: Link fibre structure to ATP-supply capacity while stating the oxygen conditions of the comparison.

Q13 / D Dialysis separates urea removal from useful-ion retention through both gradients and permeability

Answer: (D) Net movement requires both membrane permeability and an appropriate gradient.

Why: Urea has a concentration difference directed from blood to initially urea-free dialysing fluid and can cross the model membrane. Matching useful small-ion concentrations avoids a comparable initial net diffusive loss of those ions. Blood cells and large plasma proteins cannot cross despite their concentration differences because the membrane excludes them. Balanced pressure also removes an imposed bulk-filtration explanation for the selective initial solute movement.

Why not the others: (A) ignores the protein permeability restriction and reverses urea movement. (B) makes an impermeant cellular component behave like a diffusible solute. (C) treats permeability alone as sufficient for a net flux, ignoring the matched ionic concentrations.

Key point: Predict dialysis from gradients and membrane selectivity together, not from concentration or molecular size alone.

Q14 / B Edges of a pulse-labelled mitotic cohort distinguish S duration from the following G2 delay

Answer: (B) S lasts 5 h and G1 lasts 4 h.

Given: The labelled cohort initially spans the entire S phase; G2 and M durations are fixed.

Formula/Principle: The earliest labelled mitoses come from cells at the end of S, whereas the latest come from cells at its beginning.

Step 1: The first entry gives

$$T_{G2} = 2 \text{ h}.$$

Step 2: The spread of entry times gives

$$T_S = 7 - 2 = 5 \text{ h}.$$

Step 3: Subtract known phases from the cycle:

$$T_{G1} = 12 - (5 + 2 + 1) = 4 \text{ h}.$$

Why: The labelled cells have different amounts of S remaining but the same G2 still ahead. The entry-time difference therefore isolates S rather than the whole interval from pulse to the last mitosis.

Why not the others: (A) includes G2 in S. (C) exchanges G2 with S. (D) omits G2 from the full-cycle subtraction.

Key point: Use the leading and trailing edges of a pulse-labelled cohort to separate consecutive phase durations.

Q15 / C Prolactin-supported milk synthesis distinguished from oxytocin-driven myoepithelial ejection

Answer: (C) Milk synthesis and milk ejection depend on different hormone-linked processes.

Why: Prolactin promotes milk production in the mammary glands. Oxytocin released during the suckling response promotes contraction of myoepithelial cells, which helps move the milk into the ducts for ejection. Inhibiting the second process can therefore impair delivery even when milk has already been made. An observed failure of ejection is not by itself evidence that the synthetic process failed.

Why not the others: (A) collapses distinct hormones and actions into one. (B) infers absent synthesis from a downstream delivery defect. (D) reverses the mechanical role of contraction in ejection.

Key point: Distinguish production of a biological material from the regulated mechanism that releases it.

Chemistry

Q16 / C Orbital orientation distinguishes sigma, pi and symmetry-cancelled overlap

Answer: (C) Orientation relative to the internuclear axis determines the overlap symmetry.

Why: End-on pz overlap concentrates bonding overlap around the z axis and has sigma symmetry. Parallel px orbitals can overlap on the two sides of that axis, giving pi symmetry. In R, the spherical s function overlaps equally with opposite-sign px lobes, so the positive and negative contributions cancel in the ideal symmetric arrangement. Orbital proximity alone therefore does not guarantee a nonzero bonding overlap.

Why not the others: (A) exchanges end-on and sideways labels and ignores cancellation. (B) makes the lateral overlap axially symmetric and assigns an uncancelled pi interaction to s–px. (D) denies the allowed pz end-on overlap while retaining the symmetry-forbidden s–px overlap.

Key point: Use orbital orientation and signs, not merely orbital names or spatial contact, to classify an overlap.

Q17 / A Signed optical rotation fixes enantiomer composition independently of complex charge and ligand denticity

Answer: (A) The positive isomer fraction is 5/8, cobalt is +3 and coordination number is six.

Given: Only two mirror-related isomers contribute, with equal and opposite pure rotations.

Formula/Principle: Signed rotations add by isomer fraction; metal charge and donor count are independent constraints.

Step 1: If the positive fraction is f,

$$60f - 60(1 - f) = 15 \Rightarrow 120f = 75 \Rightarrow f = 5/8.$$

Step 2: Neutral en contributes no charge, so the +3 complex charge is the cobalt oxidation state.

Step 3: Three bidentate ligands provide

$$3 \times 2 = 6$$

donor atoms.

Why: Optical activity measures the imbalance between the two isomers rather than the number of ligand molecules or the metal charge. The remaining fraction 3/8 is the negative isomer, not an achiral species. Charge and denticity require separate accounting.

Why not the others: (B) uses the negative fraction and counts ligands instead of donor atoms. (C) assigns cobalt +2 despite the neutral ligands. (D) treats the measured rotation as a simple fraction of +60 without including the negative contribution.

Key point: Use signed mixture contributions, then determine oxidation state and coordination number independently.

Q18 / B Endothermic ionic dissolution can have favourable Gibbs energy after the entropy term is included

Answer: (B) Dissolution is endothermic in enthalpy yet favourable in this standard Gibbs comparison.

Given: Lattice separation is positive and hydration is negative.

Formula/Principle: Add lattice-separation and hydration enthalpies, then use delta G=delta H−T delta S.

Step 1: Add the two enthalpy steps:

$$\Delta H = 800 - 760 = 40 \text{ kJ mol}^{-1}.$$

Step 2: Convert entropy units:

$$T\Delta S = 298(0.150) = 44.7 \text{ kJ mol}^{-1}.$$

Step 3:

$$\Delta G = 40 - 44.7 = -4.7 \text{ kJ mol}^{-1}.$$

Why: Hydration does not quite repay lattice separation in enthalpy, but the favourable entropy term exceeds that residual energy cost. This is a standard-state comparison; it does not assert unlimited dissolution at every concentration.

Why not the others: (A) reverses the net enthalpy sign. (C) adds hydration release as a positive cost. (D) adds rather than subtracts T delta S.

Key point: Competition between lattice and hydration enthalpies is separate from the entropy contribution to spontaneity.

Q19 / D Photon absorption and electron-impact excitation impose different constraints on excess incident energy

Answer: (D) Discrete photon absorption differs from an inelastic collision with a surviving projectile.

Why: Absorbing one photon transfers its whole energy, so a bound-state transition must match an allowed level gap under the stated idealisations. Twelve electronvolts matches neither listed excitation and is below ionisation. A colliding electron can instead lose exactly 10.2 eV while retaining the remaining 1.8 eV as kinetic energy. It cannot reach the 12.09 eV level with only 12.0 eV initially. Thus quantisation constrains the energy deposited in the atom, not necessarily the entire initial energy of a material projectile.

Why not the others: (A) keeps part of a photon after claiming its absorption. (B) confuses excitation with removal of the electron. (C) ignores discrete excited states below the ionisation threshold and the outgoing electron's energy.

Key point: Apply quantised atomic energy transfer with the correct fate of the incident probe.

Q20 / A Sodium-fusion halogen analysis requires removal of cyanide and sulfide without adding chloride

Answer: (A) Nitric-acid treatment removes interfering fusion products without introducing the chloride sought.

Why: Sodium fusion converts covalently bound elements into ionic forms suitable for aqueous testing. Cyanide and sulfide can form silver precipitates and must be removed by the prescribed nitric-acid boiling treatment before interpreting the halogen test. Hydrochloric acid would introduce chloride itself, creating a false-positive source of AgCl. Testing an untreated covalent organic chloride as though its chlorine were already a free ion would also bypass the purpose of fusion.

Why not the others: (B) introduces rather than removes chloride. (C) denies the silver-salt interference of the other fusion products. (D) assumes covalent bonding gives the same aqueous behaviour as ionic chloride.

Key point: Convert the analyte to the tested ionic form and eliminate both interferents and reagent-derived contamination.

Q21 / B Catalyst deactivation permits acyl-chloride reduction to stop at an oxidisable aldehyde

Answer: (B) Controlled catalytic reduction preserves the aldehyde stage.

Why: Rosenmund reduction converts an acyl chloride to the corresponding aldehyde without changing the carbon skeleton. Palladium activity is reduced to limit further hydrogenation of the newly formed aldehyde. The desired benzaldehyde therefore retains its carbonyl and aldehydic hydrogen and can undergo aldehyde oxidation. The support and catalyst control do not introduce a new nitrogen or oxygen atom into the product.

Why not the others: (A) reverses the intended chemoselectivity and destroys the retained aromatic skeleton. (C) mistakes a reducing hydrogen environment for an oxidation to carboxylic acid. (D) invents nitrogen transfer from a sulfate support containing no nitrogen.

Key point: A preparation may require suppression of a later reaction as well as promotion of the first one.

Q22 / C Isotope tracking distinguishes nitrogen-retaining azo coupling from nitrogen-losing Sandmeyer substitution

Answer: (C) Coupling retains the diazonium nitrogen pair, whereas Sandmeyer replacement ejects it.

Step 1: In azo coupling, the activated phenoxide ring bonds to the terminal nitrogen of the diazonium group, giving an aryl–N=N–aryl linkage after work-up. Both labelled nitrogen atoms remain in that linkage.

Step 2: With CuCl, the diazonium group is replaced by chlorine and nitrogen is evolved; both labels therefore leave in N2.

Why: The same starting diazonium salt can act as a coupling electrophile or as a precursor to substitution with loss of nitrogen. Product colour and connectivity arise from the retained azo conjugation in P, not from conversion to an ordinary amine. Label conservation must follow the actual route.

Why not the others: (A) transfers the nitrogen-loss rule to the coupling route. (B) invents amine and nitrile products unsupported by the reagents. (D) exchanges the preparations and proposes impossible attachment of a nitrogen label to the chlorine atom.

Key point: A common reactive intermediate can retain or eject the same atoms through different reaction pathways.

Q23 / A Mixture nucleotide accounting separates a duplex base constraint from single-stranded RNA composition

Answer: (A) DNA supplies 40% of nucleotide units, and the single-stranded component breaks aggregate G=C.

Given: Only DNA contributes thymine in the stated mixture.

Formula/Principle: Conserve nucleotide units and apply complementary-base equality only to the duplex component.

Step 1: Let its unit fraction be f:

$$20f = 8 \Rightarrow f = 0.40.$$

Step 2: Weight the guanine and cytosine fractions separately:

$$G = 0.40(30) + 0.60(35) = 33, \quad C = 0.40(30) + 0.60(25) = 27.$$

Why: Complementary-base equality applies to the complete double-stranded DNA component, not automatically to a separate single RNA strand. Hydrolysis mixes units without creating new base pairs, so its total composition retains the RNA asymmetry. The inferred fraction counts nucleotide units, not the numbers of original polymer molecules.

Why not the others: (B) extends a duplex constraint to single-stranded RNA. (C) exchanges DNA and RNA fractions. (D) invents a required T=U relation between different components.

Key point: Apply complementary-base accounting to the component whose strand structure justifies it.

Q24 / D Parallel Arrhenius plots separate equal barriers from different pre-exponential factors

Answer: (D) Equal slopes imply equal activation energies, while the vertical separation compares prefactors.

Given:

$$\ln k = \ln A - E_a/(RT).$$

Formula/Principle: In an Arrhenius plot the slope is −Ea/R and the intercept is ln A.

Step 1: The slope against 1/T is −Ea/R; parallel slopes therefore give equal Ea.

Step 2: With that term cancelling,

$$\ln k_u - \ln k_l = \ln(A_u/A_l) = \ln 100,$$

so Au/Al=100.

Why: A vertical change at unchanged slope is a prefactor change, which can reflect differences such as effective collision orientation. A larger rate constant at one temperature alone would not isolate the cause, but the shared slope supplies the needed comparison.

Why not the others: (A) assigns the intercept ratio to the slope. (B) reads height as barrier without checking slope. (C) ignores the nonzero vertical separation.

Key point: Read temperature sensitivity from the slope and multiplicative rate scale from the intercept.

Q25 / B Tracer incorporation demonstrates dynamic chemical equilibrium despite constant bulk concentrations

Answer: (B) Tracer exchange reveals continuing microscopic reactions within a macroscopically stationary equilibrium.

Why: A labelled nitrogen atom can enter ammonia only through chemical conversion, so some forward reaction continues. In a closed equilibrated system with unchanged conditions and effectively constant composition, that conversion is balanced on average by reverse reaction. The tiny tracer is assumed not to impose a significant composition shift, and its isotope label tracks atoms rather than creating new atoms. Equilibrium therefore concerns equal opposing rates, not molecular inactivity.

Why not the others: (A) cannot sustain the stated closed-system composition while conversion proceeds. (C) assigns a changed equilibrium constant to a catalyst, which accelerates approach without changing that constant. (D) mistakes tracing an atom into a new molecule for loss of atom conservation.

Key point: Distinguish a stationary bulk composition from a frozen molecular population.

Q26 / D Peroxide disproportionation has internal electron transfer without a required external circuit

Answer: (D) Oxygen gas is 0.0500 mol and internal redox transfer is 0.100 mol electron equivalents.

Given: Peroxide oxygen is −1; product oxygen is −2 in water and zero in O2.

Formula/Principle: Conserve atoms and equate peroxide oxygen oxidation and reduction electron equivalents.

Step 1: Stoichiometry gives

$$n(\mathrm{O_2}) = 0.100/2 = 0.0500 \text{ mol}.$$

Step 2: Two peroxide oxygen atoms forming each O2 each lose one electron equivalent. Thus

$$n_e = 2n(\mathrm{O_2}) = 0.100 \text{ mol}.$$

Step 3: The same number is accepted by oxygen atoms reduced to water.

Why: Disproportionation contains oxidation and reduction of the same initial oxidation state. Electron-equivalent bookkeeping describes that internal chemical transfer; it does not imply free electrons travel through a missing external wire.

Why not the others: (A) omits the two-peroxide requirement per O2. (B) equates no external current with no oxidation-state change. (C) halves the gas amount twice.

Key point: Keep product stoichiometry, internal electron equivalents and external electrical current distinct.

Q27 / C Reservoir entropy and system entropy compete to reverse an exothermic reaction's feasibility with temperature

Answer: (C) Reservoir entropy overcomes the system entropy loss at 400 K but not at 600 K.

Given: At constant pressure the system releases 48000 J per mole to its reservoir.

Formula/Principle: Reservoir entropy is heat received divided by its temperature; delta G=delta H−T delta S.

Step 1: At 400 K,

$$\Delta S_{surr} = 48000/400 = 120 \ \mathrm{J\,K^{-1}}, \quad \Delta S_{tot} = 120 - 100 = 20 \ \mathrm{J\,K^{-1}}.$$

Step 2:

$$\Delta G_{400} = -48 - 400(-0.100) = -8 \ \mathrm{kJ\,mol^{-1}}.$$

Step 3: At 600 K,

$$\Delta G_{600} = -48 - 600(-0.100) = 12 \ \mathrm{kJ\,mol^{-1}}.$$

Why: Exothermic heat release raises surroundings entropy, while the negative system entropy resists this direction. Increasing temperature weakens the surroundings entropy gain per unit released heat and strengthens the Gibbs penalty from system entropy loss. The sign changes at 480 K under the stated approximation.

Why not the others: (A) reverses the total sign. (B) adds the magnitude of a negative system entropy. (D) omits the entropy competition.

Key point: Evaluate the system and reservoir contributions with signs before deciding whether heat release is sufficient for spontaneity.

Q28 / B Mass conservation and measured density determine concentration without assuming additive solution volumes

Answer: (B) Conserved solute mass is 34 g, while final volume follows from final density.

Given: Both initial solution volumes, densities and mass fractions are specified, and final density is measured.

Formula/Principle: Conserve solution and solute masses, then infer the final volume from measured density.

Step 1: Initial solution masses are 120 g and 100 g; their solute masses are

$$0.20(120) = 24 \ \mathrm{g}, \quad 0.10(100) = 10 \ \mathrm{g}.$$

Step 2: Total mass is 220 g and solute amount is 34/100=0.340 mol. Final volume is

$$V = 220/1.25 = 176 \ \mathrm{mL} = 0.176 \ \mathrm{L}.$$

Step 3:

$$M = 0.340/0.176 = 85/44 \ \mathrm{mol\,L^{-1}}, \quad w\% = 100(34/220) = 170/11\%.$$

Why: Mass is conserved but the measured density explicitly determines a non-additive final volume. Mass percentage must be weighted by solution masses, not averaged as two equal volumes.

Why not the others: (A) assumes both additive volumes and an unweighted mass-percent average. (C) adds percentages instead of solute masses. (D) calculates mass fraction correctly but still uses 200 mL.

Key point: Use separate conservation and density steps when mixing solutions with unequal densities and volume contraction.

Q29 / A Hydrogenation stoichiometry and mild bromine response distinguish aromatic from ordinary cyclic unsaturation

Answer: (A) Aromatic stability changes the ordinary bromine response without removing benzene's hydrogenation demand.

Why: Converting benzene to cyclohexane adds six hydrogen atoms, corresponding to three H2 molecules. Cyclohexene requires only one H2 to saturate its one ordinary double bond. Under the specified mild bromine-test conditions, cyclohexene undergoes addition, whereas benzene does not behave like three isolated alkene bonds. Benzene's aromatic delocalisation makes loss of aromaticity by simple addition comparatively unfavourable.

Why not the others: (B) treats aromatic bonding as three independent ordinary alkenes in the bromine test and overcounts cyclohexene hydrogenation. (C) exchanges the formula changes. (D) confuses a ring with saturation.

Key point: Separate formula-based hydrogen demand from the reaction behaviour conferred by aromatic stabilisation.

Q30 / D Branching-dependent boiling point and carbocation-controlled solvolysis give independent isomer rankings

Answer: (D) Intermolecular attractions and carbocation-forming kinetics are different comparisons.

Why: The isomers have equal molar mass, so their boiling-point difference is related to molecular shape and effective contact in the liquid rather than an atomic-mass difference. Branching can reduce the contact responsible for dispersion attractions. The separate SN1 comparison instead depends strongly on the relative stability of a tertiary versus primary carbocation and the ease of ionisation in the solvent. A molecule can therefore boil lower and react faster without any contradiction.

Why not the others: (A) infers an intermediate's stability from a bulk phase-change temperature. (B) gives boiling a chemical solvolysis transition state. (C) denies the shared molecular formula and does not address carbocation formation.

Key point: Identify whether the observation concerns intermolecular cohesion or a chemical reaction pathway before choosing its explanation.

Mathematics

Q31 / B A focal distance ratio fixes eccentricity before completing the square locates the centre

Answer: (B) The eccentricity is 1/2 and the ellipse centre is (2/3,0).

Given: At the origin the focus distance is 1 and directrix distance is 2.

Formula/Principle: The eccentricity is the focus/directrix distance ratio; completing the locus square gives its centre.

Step 1:

$$1 = 2e \Rightarrow e = 1/2.$$

Step 2: Square the distance-ratio equation:

$$(x-1)^2 + y^2 = \tfrac{1}{4}(x-2)^2.$$

Step 3: Simplification gives

$$\tfrac{3}{4}x^2 - x + y^2 = 0,$$

whose x-coordinate after completing the square is 2/3.

Worked derivation:

$$\tfrac{3}{4}(x - 2/3)^2 + y^2 = 1/3.$$

Why: The focus is not generally the centre of a conic. The given point fixes eccentricity, and the resulting locus equation fixes the translation independently.

Why not the others: (A) inverts the distance ratio. (C) mistakes focus for centre. (D) ignores the ratio established at the origin.

Key point: Use a known point to determine the ratio, then recover the geometry from the complete locus.

Q32 / D Opposite tangent and normal intercepts survive an absolute-value branch check

Answer: (D) x^4=1/3.

Given: The tangent slope is $-1/x^2$, so the normal slope is x^2.

Formula/Principle: Find the tangent and perpendicular normal intercepts, then equate their absolute lengths.

Step 1: Setting y=0 in the tangent equation gives its x-intercept

$$t = 2x.$$

Step 2: The normal's x-intercept is

$$n = x - 1/x^3.$$

Step 3: Equal distances require |n|=2x. The case n=2x gives $-1/x^3=x$, impossible for x>0. The other case gives

$$x - 1/x^3 = -2x \Rightarrow 3x^4 = 1.$$

Worked derivation:

$$x = 3^{-1/4}, \quad n = -2x, \quad t = 2x.$$

Why: Equal intercept lengths do not require the intercepts to have the same sign. Resolving the absolute-value cases is essential to retaining the valid opposite-side arrangement.

Why not the others: (A) inverts the final fourth-power relation. (B) forces the normal intercept to zero. (C) omits one x contribution when equating opposite intercepts.

Key point: Translate equal geometric lengths into absolute values before choosing signs.

Q33 / C A strict shape constraint turns a familiar area maximum into an unattained supremum

Answer: (C) The area has supremum 9, but no admissible rectangle attains it.

Given: If the length is x, the width is 6−x. Positivity and strict ordering give 3<x<6.

Formula/Principle: Optimise on the actual open domain, not its closed extension.

Step 1:

$$A = x(6 - x) = 9 - (x - 3)^2.$$

Step 2: Thus A<9 for every admissible x, while values approach 9 as x decreases to 3.

Worked derivation:

$$\sup_{3<x<6} A = 9, \quad A = 9 \iff x = 3,$$

and that square is excluded.

Why: A limiting best value need not be an attained maximum. The familiar fixed-perimeter square result identifies the boundary of the admissible set here rather than a legal optimum.

Why not the others: (A) calculates the sample rectangle's area incorrectly and does not maximise. (B) ignores the strict inequality. (D) confuses perimeter with area and chooses the vanishing-area boundary.

Key point: Check whether an extremising point belongs to the physical domain before calling a bound a maximum.

Q34 / A Factor cancellation, one-point redefinition and absolute-value sign splitting affect different parts of an integral

Answer: (A) The integral is pi−2 arctan 2.

Given: The exceptional value at one point does not change the integral of this bounded piecewise-continuous function.

Formula/Principle: Factor the rational expression, split where its sign changes, and disregard a bounded one-point redefinition in integration.

Step 1: Factor

$$x^3 - x^2 - x + 1 = (x-1)^2(x+1),$$

so for x≠1,

$$f(x) = \frac{x^2-1}{x^2+1} = 1 - \frac{2}{x^2+1}.$$

Step 2: This is negative on [0,1) and positive on (1,2]. A primitive is F(x)=x−2 arctan x.

Step 3: Split the absolute-value integral:

$$I = -[F(1) - F(0)] + [F(2) - F(1)].$$

Worked derivation:

$$I = -(1 - \pi/2) + (2 - 2\arctan 2) - (1 - \pi/2) = \pi - 2\arctan 2.$$

Why: Cancellation removes the algebraic denominator zero away from the specified point; it does not make the chosen value 7 the continuous extension. Nevertheless an isolated value contributes no interval width or area.

Why not the others: (B) integrates the signed function without the sign split. (C) keeps only the arctangent part with the wrong sign. (D) treats a single point's height as a unit-width area.

Key point: Distinguish function values at isolated points from contributions over intervals.

Q35 / C Commutation propagates finite-function images along a cycle but permits a nonbijective collapse

Answer: (C) There are four commuting maps, three of them bijections.

Given: f has a three-cycle and exactly one fixed point, 4.

Formula/Principle: Commutation propagates images along each orbit, and maps fixed points to fixed points.

Step 1: Commutation at 4 gives

$$g(4) = f(g(4)),$$

so g(4)=4.

Step 2: Choose g(1)=a. Then commutation forces

$$g(2) = f(a), \quad g(3) = f^2(a).$$

Each of the four choices a∈S is consistent since f³(a)=a.

Step 3: If a is 1,2 or 3, the first three images permute that cycle and g is bijective. If a=4, all four elements map to 4.

Worked derivation:

$$N_g = 4, \quad N_{\text{bijective}} = 3.$$

Why: The images are constrained by the orbit structure, not independently chosen four times. Commutation permits collapse of a whole cycle onto a fixed point, so it does not by itself imply invertibility.

Why not the others: (A) discards the constant map. (B) calls the collapsed map a bijection. (D) ignores the forced cycle images and fixed-point condition.

Key point: Determine an image on each orbit, then propagate the composition constraint.

Q36 / B An oriented cross-product equation couples perpendicular components and removes the parallel component

Answer: (B) v=e2−e3.

Given: The basis is orthonormal and its orientation fixes the cross-product signs.

Formula/Principle: Resolve the vector equation in a right-handed orthonormal basis.

Step 1: Write v=a e1+b e2+c e3. Then

$$e_1 \times v = be_3 - ce_2.$$

Step 2: Comparing basis components gives

$$a = 0, \quad b - c = 2, \quad c + b = 0.$$

Thus b=1,c=−1.

Worked derivation:

$$(e_2 - e_3) + e_1 \times (e_2 - e_3) = (e_2 - e_3) + (e_3 + e_2) = 2e_2.$$

Why: The parallel component cannot be supplied by the cross product, while the two perpendicular components are coupled with orientation-dependent signs. Direct substitution checks both the scale and handedness.

Why not the others: (A) reverses the e3 sign. (C) gives twice the required right-hand side. (D) leaves a nonzero e1 component absent from the target.

Key point: Resolve a vector equation in an oriented basis rather than treating cross multiplication as scalar division.

Q37 / D A nilpotent matrix part produces an exponent factor before an entry ratio fixes the power

Answer: (D) n=6 and the trace is 128.

Given: Write A=2I+N with N having only its upper-right entry equal to 1, so $N^2=0$.

Formula/Principle: In the binomial expansion of commuting 2I and N, all powers N^2 and higher vanish.

Step 1:

$$A^n = 2^n I + n2^{n-1} N.$$

Step 2: The entry condition gives

$$n2^{n-1} = 3(2^n) \Rightarrow n = 6.$$

Step 3: Both diagonal entries are 2^6=64.

Worked derivation:

$$\text{tr}(A^6) = 2(64) = 128.$$

Why: A repeated diagonal value does not make A a scalar matrix; the nilpotent part leaves a term proportional to n. That term must be retained before comparing the entries.

Why not the others: (A) omits the factor 1/2 in the off-diagonal/diagonal ratio. (B) gives one diagonal entry instead of the trace. (C) has the right trace attached to the wrong exponent.

Key point: Repeated-factor matrix identities can preserve a polynomial factor in the power index.

Q38 / A A sum-threshold stopping report reweights the first member of an ordered sample without replacement

Answer: (A) The conditional probability is 1/3.

Given: Every ordered pair of distinct first two numbers has the same probability, and no possible first number reaches 6 alone.

Formula/Principle: Condition ordered samples on the complete stopping event, not on one arbitrarily labelled outcome.

Step 1: Stopping on the second reveal is therefore equivalent to a+b≥6 with a≠b.

Step 2: For first values 1,2,3,4,5 respectively, the allowable second-value counts are

$$1, 2, 2, 3, 4.$$

Step 3: There are 12 qualifying ordered pairs, four beginning with 5.

Worked derivation:

$$P(a = 5 \mid \text{stop at } 2) = \frac{4}{1+2+2+3+4} = \frac{1}{3}.$$

Why: The stopping report favours larger first values because more second values then meet the threshold. The original equal first-value probabilities cannot be retained after conditioning on the stopping time.

Why not the others: (B) ignores that selection effect. (C) treats the two reveal positions as two equally likely first-value outcomes. (D) divides by a subset of the qualifying pairs instead of the whole conditional event.

Key point: Translate the stopping rule into complete constraints on the revealed ordered sample.

Q39 / C Exactly one complete couple is counted separately from two different partial couples

Answer: (C) There are 240 committees.

Given: The complete couple is unique in every admissible committee.

Formula/Principle: Select the unique complete couple, then select two distinct partial couples and one partner from each.

Step 1: Choose that complete couple in 6 ways.

Step 2: Choose two of the other five couples to supply one member each:

$$\binom{5}{2} = 10.$$

Step 3: Choose either partner from each of those two couples, giving 2^2=4 choices.

Worked derivation:

$$N = 6\binom{5}{2}\,2^2 = 6(10)(4) = 240.$$

Why: Selecting two different remaining couples prevents an unintended second complete couple. The unique fully selected couple ensures each committee is counted once, and committee members have no order.

Why not the others: (A) orders the two partial couples unnecessarily. (B) misses one binary partner choice. (D) includes incompatible or extra couple selections rather than the stated one-complete/two-partial structure.

Key point: Identify a uniquely determined structural component before multiplying independent choices.

Q40 / D A finite geometric identity creates a coefficient gap and a shifted binomial boundary

Answer: (D) The difference is 4.

Given: The alternating geometric polynomial ends with the positive x^10 term.

Formula/Principle: Use its exact finite identity before extracting boundary coefficients.

Step 1:

$$(1+x)(1-x+x^2-\cdots+x^{10})=1+x^{11}.$$

Step 2: Consequently

$$P(x)=(1+x)^4(1+x^{11}).$$

Step 3: There is no x^10 term. The x^12 term comes only from x^11 times the x term of (1+x)^4, with coefficient 4.

Worked derivation:

$$[x^{12}]P-[x^{10}]P=\binom{4}{1}-0=4.$$

Why: Finite cancellation leaves a shifted low-degree block rather than an infinite geometric series. The high-degree coefficient is supplied by that shifted block, while the intervening degrees vanish.

Why not the others: (A) reverses the requested subtraction. (B) extends the gap to the shifted block. (C) extracts from the uncancelled fifth power without reducing it.

Key point: Preserve the finite endpoint term when cancelling a geometric-polynomial factor.

Q41 / A A positive endpoint geometric mean leaves a signed GP branch that the sum and ratio condition resolve

Answer: (A) The ratio is −2 and the four-term sum is −15.

Given: ac=36 and the GP relation gives b²=ac, not necessarily b=+sqrt(ac).

Formula/Principle: Use b²=ac for a real GP while retaining both signs of b; use sum and ratio constraints to choose the branch.

Step 1: The sum conditions give b=9−15=−6.

Step 2: a,c are roots of

$$t^2-15t+36=0\Rightarrow t=3,12.$$

Step 3: If a=3,c=12, r=b/a=−2. If a=12,c=3, r=−1/2. The magnitude condition selects the first order.

Step 4: The fourth term is 12(−2)=−24.

Worked derivation:

$$S_4=3-6+12-24=-15.$$

Why: The conventional positive geometric mean of a and c does not force the intervening signed GP term to be positive. The sum fixes that sign, while ordering requires the separate ratio condition.

Why not the others: (B) discards the negative middle term. (C) retains the wrong ordering. (D) makes the fourth term positive despite the negative ratio.

Key point: Separate a positive geometric mean from a signed term of a real geometric progression.

Q42 / C Conjugate-root modulus fixes a coefficient before a discriminant and quadrant check

Answer: (C) a=1 and the positive-imaginary root has argument pi/3.

Given: For conjugate roots z and conjugate(z), their product equals |z|²=4.

Formula/Principle: For conjugate roots their product equals the squared modulus; verify the discriminant afterwards.

Step 1: Vieta gives

$$a + 3 = 4 \Rightarrow a = 1.$$

Step 2: The equation becomes x²−2x+4=0, with discriminant −12, confirming the required nonreal roots.

Step 3: Its positive-imaginary root is

$$z = 1 + i\sqrt{3} = 2(\cos(\pi/3) + i\sin(\pi/3)).$$

Worked derivation:

$$|z| = 2, \quad \operatorname{Arg} z = \pi/3.$$

Why: The modulus condition determines the constant term only because conjugacy converts the root product to a positive squared modulus. Checking the discriminant prevents applying the same interpretation to an inadmissible real-root parameter.

Why not the others: (A) violates the required product. (B) places a positive-real-part root in the wrong quadrant. (D) changes both product and root geometry.

Key point: Use conjugacy to connect Vieta's product with modulus, then verify the root type and argument.

Q43 / B A mean constraint and asymmetric bounds determine an attainable maximum mean deviation

Answer: (B) The greatest mean deviation is 12/5.

Given: The total of the five observations is zero.

Formula/Principle: Bound absolute value by the line joining its endpoint values on the permitted interval.

Step 1: For every −2≤x≤3,

$$|x| \le x/5 + 12/5.$$

For x≥0 this is equivalent to x≤3; for x≤0 it is equivalent to x≥−2.

Step 2: Average over the five values:

$$\frac{1}{5}\sum |x_i| \le \frac{1}{25}\sum x_i + 12/5 = 12/5.$$

Step 3: The data −2,−2,−2,3,3 have mean zero and attain the bound.

Worked derivation:

$$MD = (2 + 2 + 2 + 3 + 3)/5 = 12/5.$$

Why: A bound is a maximum only after an admissible dataset attains it. The prescribed sample size permits exactly the endpoint balance needed here.

Why not the others: (A) overlooks the allowed positive endpoint contributions. (C) treats all absolute deviations as 3 despite the mean and negative bound. (D) exceeds the proved pointwise bound after averaging.

Key point: Prove a universal dispersion bound and exhibit a dataset satisfying every constraint.

Q44 / A A finite solution limit does not permit crossing an excluded logarithmic coefficient singularity

Answer: (A) The solution is 2 ln x on (1,infinity), and x=1 remains excluded.

Given: The coefficient 1/(x ln x) is undefined at x=1.

Formula/Principle: On a nonzero branch, separate dy/y=dx/(x ln x); also check the equilibrium branch y=0.

Step 1: Integrating gives

$$\ln|y| = \ln(\ln x) + C, \quad y = K \ln x \quad (x > 1).$$

Step 2: Initial data fix

$$2 = K \ln e \Rightarrow K = 2.$$

Step 3: Direct substitution verifies y'=2/x and y/(x ln x)=2/x for every x>1. The zero solution does not satisfy y(e)=2.

Worked derivation:

$$\lim_{x \downarrow 1} 2 \ln x = 0, \quad \text{but } 1/(x \ln x) \text{ is undefined at } 1.$$

Why: The curve can have a finite boundary value even when the original differential equation cannot be continued through that boundary. Cancelling ln x after inserting the solution does not repair the original domain.

Why not the others: (B) fails substitution. (C) crosses the excluded singular point. (D) reverses the logarithmic dependence and derivative sign.

Key point: Verify a solution in the original equation and retain the original coefficient domain.

Q45 / D A restricted piecewise inverse has a smooth interior join but an unbounded endpoint derivative

Answer: (D) The restricted monotone pieces give a continuous inverse with a regular join but singular endpoint slope.

Given: Both pieces of f are increasing on their stated domains, meet at value 1 and together cover all nonnegative outputs.

Formula/Principle: Invert each monotone image interval, then compare one-sided values and difference quotients.

Step 1: Invert each image interval:

$$g(y) = \sqrt{y} \ (0 \le y \le 1), \quad g(y) = (y+1)/2 \ (y > 1).$$

Step 2: The two values and derivative limits at 1 agree:

$$g(1) = 1, \quad g'_-(1) = 1/2 = g'_+(1).$$

Step 3: At zero, the right difference quotient is

$$(\sqrt{h} - 0)/h = 1/\sqrt{h} \to \infty.$$

Worked derivation: Thus g'(1)=1/2, while no finite right derivative exists at zero.

Why: Invertibility depends on the restricted domain actually given, not the behaviour of x^2 on negative inputs. A smooth join can coexist with a different endpoint's unbounded inverse slope.

Why not the others: (A) ignores equality of the two branch limits. (B) confuses endpoint value with derivative and misses the reciprocal slope. (C) imports excluded negative inputs.

Key point: Check inverse image intervals, join limits and endpoint derivatives separately.

Physics

Q46 / A A movable pulley combines two supporting tensions with a two-to-one acceleration constraint

Answer: (A) The supported load rises at 10/11 m s−2 and the hanging mass falls twice as fast.

Given: Two rope segments support the movable pulley; let its upward acceleration be a.

Step 1: Rope length conservation gives downward acceleration 2a for the other mass.

Step 2: Newton's equations are

$$2T - 30 = 3a, \quad 20 - T = 2(2a).$$

Step 3: From T=20−4a,

$$40 - 8a - 30 = 3a \Rightarrow a = 10/11 \text{ m s}^{-2}.$$

Worked derivation:

$$T = 180/11 \text{ N}, \quad a_{2kg} = 20/11 \text{ m s}^{-2} \text{ downward}.$$

Why: Two supporting tensions and a two-to-one motion constraint replace the equal-acceleration model of a fixed-pulley Atwood machine. The positive solution validates the assumed directions.

Why not the others: (B) uses equal accelerations and a direct two-mass weight difference. (C) ignores the rope's factor-two constraint. (D) compares masses without their mechanical advantage.

Key point: Derive the rope constraint before applying force balance to a movable pulley.

Q47 / B A speed-dependent power law cancels linear drag to produce constant net acceleration

Answer: (B) The final speed is 8.0 m s−1 and acceleration is constant at 3.0 m s−2.

Given: Power is specified as a function of positive speed, not held constant.

Formula/Principle: At positive speed, engine force is power divided by speed; subtract drag before applying Newton law.

Step 1: Convert engine power to driving force:

$$F_{eng} = P/v = v + 6 \text{ N}.$$

Step 2: Subtract drag:

$$F_{net} = (v + 6) - v = 6 \text{ N}, \quad a = 6/2 = 3 \text{ m s}^{-2}.$$

Step 3: Use constant-acceleration work/kinematics:

$$v_f^2 = 2^2 + 2(3)(10) = 64.$$

Worked derivation: At v=8,

$$P_{eng} = 112 \text{ W}, \; P_{drag} = 64 \text{ W}, \; P_{net} = 48 \text{ W} = mav.$$

Why: Speed-dependent engine and drag contributions cancel in force, even though neither engine power nor drag power is constant. The positive-speed assumption permits division by v.

Why not the others: (A) misses the work of the net force. (C) incorrectly calls the given power constant. (D) divides by the mass twice.

Key point: Convert power to force and combine forces before choosing the acceleration model.

Q48 / C Two shell-potential measurements reconstruct masses while a nonzero constant interior potential gives zero work

Answer: (C) The masses are 4M and 3M, and internal-point displacement requires zero net gravitational work.

Given: Inside a thin spherical shell its potential is constant at −Gm/radius, although it is generally nonzero.

Formula/Principle: A shell gives constant interior potential and point-mass exterior potential; work uses potential differences.

Step 1: At the centre,

$$m_1 + m_2/3 = 5M.$$

Step 2: At 2R, the inner shell acts as a central mass while the outer potential stays constant:

$$m_1/2 + m_2/3 = 3M.$$

Step 3: Subtracting gives m1/2=2M, hence m1=4M and m2=3M.

Worked derivation: Inside R,

$$V = -G(4M)/R - G(3M)/(3R) = -5GM/R, \quad W_{ext} = m_{test}\Delta V = 0.$$

Why: The field is the spatial variation of potential; a constant negative potential gives zero field and zero potential difference within that region. Both shell contributions must be retained in the inverse mass calculation.

Why not the others: (A) fits the central value but not the 2R value. (B) confuses potential value with potential change. (D) ignores the second measured value and positivity of both masses.

Key point: Use the shell theorem for each region and compute work from differences, not absolute potential.

Q49 / D Stress–strain hysteresis separates returned energy, dissipation and residual deformation

Answer: (D) Dissipation is 1.0 J and residual strain is 0.005.

Given: Work per initial volume is the signed area under the stress–strain path.

Step 1: Loading work density is

$$u_{in} = \tfrac{1}{2}(0.020)(200 \times 10^6) = 2.0 \times 10^6 \ \mathrm{J\,m^{-3}}.$$

Step 2: Unloading returns

$$u_{out} = \tfrac{1}{2}(0.020 - 0.005)(200 \times 10^6) = 1.5 \times 10^6 \ \mathrm{J\,m^{-3}}.$$

Step 3: Multiply their difference by volume:

$$E_{diss} = 0.5 \times 10^6 (2.0 \times 10^{-6}) = 1.0 \ \mathrm{J}.$$

Worked derivation:

$$E_{in} = 4.0 \ \mathrm{J}, \quad E_{out} = 3.0 \ \mathrm{J}, \quad E_{in} - E_{out} = 1.0 \ \mathrm{J}.$$

Why: The zero-stress intercept shows permanent deformation; unloading area, rather than total loading area, measures recovered mechanical energy. The remaining path can close at zero stress without additional work.

Why not the others: (A) calls all input dissipated and misses residual deformation. (B) reports recovered energy. (C) gets the loop energy but ignores its shifted intercept.

Key point: Separate input work, recoverable work and permanent strain on a hysteretic path.

Q50 / B Opposed free expansions cancel when coefficient–length products match, not when lengths match

Answer: (B) The lengths must be 2.0 m and 1.0 m.

Given: The signed geometry is a difference of freely expanding lengths, not a bonded stress constraint.

Formula/Principle: Differentiate the signed length relation with respect to a common temperature.

Step 1: The reference separation requires

$$L_1 - L_2 = 1.$$

Step 2: Its thermal change is

$$\Delta X = (\alpha_1 L_1 - \alpha_2 L_2)\Delta T.$$

Step 3: Compensation requires L1=2L2; combining the equations gives L2=1 m,L1=2 m.

Worked derivation:

$$\alpha_1 L_1 = 10^{-5}(2) = 2 \times 10^{-5}(1) = \alpha_2 L_2.$$

Why: The two coefficients need not match; equal absolute expansions cancel because the mechanism subtracts their displacements. The model neglects higher-order expansion and temperature differences.

Why not the others: (A) reverses both separation and compensation. (C) satisfies separation but not equal expansions. (D) gives zero separation and unequal expansions.

Key point: Apply the signed mechanical geometry before setting a thermal compensation condition.

Q51 / D Per-particle collision frequency and volume event rate respond differently to density–speed compensation

Answer: (D) Per-molecule collision frequency is unchanged, while event density doubles.

Given: Characteristic speed is proportional to sqrt(T), and cross-section is fixed.

Formula/Principle: Per-molecule collision frequency scales as n sigma v; pair events per volume scale as n times that frequency divided by two.

Step 1: Speed changes by

$$\bar{v}_2/\bar{v}_1 = \sqrt{1/4} = 1/2.$$

Step 2: Per-molecule frequency scales as n sigma v, so

$$z_2/z_1 = 2(1/2) = 1.$$

Step 3: Each event involves two molecules, giving event density nz/2. Thus

$$\mathcal{R}_2/\mathcal{R}_1 = 2(1) = 2.$$

Worked derivation: Equivalently,

$$\mathcal{R} \propto n^2 \bar{v} \Rightarrow \mathcal{R}_2/\mathcal{R}_1 = 4/2 = 2.$$

Why: A molecule sees unchanged collision opportunities per time because greater density compensates lower speed. There are nevertheless twice as many molecules in each volume, so the total event count rises. The factor one-half prevents double-counting each pair collision.

Why not the others: (A) transfers a per-particle result to a per-volume count. (B) ignores the speed decrease. (C) ignores the increased target density.

Key point: State whether a rate is per object or per volume before applying kinetic scaling.

Q52 / A Two displacement–speed observations determine frequency before total SHM energy is reconstructed

Answer: (A) Total mechanical energy is 0.025 J.

Given: Both observations belong to the same oscillator and its energy is constant.

Formula/Principle:

$$v^2 = \omega^2(A^2 - x^2).$$

Step 1: Subtract the observations:

$$0.40^2 - 0.30^2 = \omega^2(0.040^2 - 0.030^2),$$

so omega²=0.070/0.000700=100 s−2.

Step 2: Recover amplitude:

$$A^2 = 0.030^2 + 0.40^2/100 = 0.00250 \text{ m}^2.$$

Step 3:

$$E = \tfrac{1}{2}m\omega^2A^2 = \tfrac{1}{2}(0.20)(100)(0.00250) = 0.025 \text{ J}.$$

Worked derivation: The first observation checks

$$K + U = 0.016 + 0.009 = 0.025 \text{ J}.$$

Why: One observation alone leaves frequency and amplitude undetermined, but subtraction eliminates the shared amplitude term. Energy then includes both kinetic and potential contributions at either observation.

Why not the others: (B) omits the factor one-half. (C) reports only the first kinetic energy. (D) reports only its potential energy.

Key point: Use two phase-space observations to eliminate an unknown shared oscillator parameter.

Q53 / C Voltage-controlled capacitor attraction creates a stability threshold beyond mere force balance

Answer: (C) Stable equilibria have x<d/3, ending at the marginal threshold with the stated voltage.

Given: The restoring spring force is kx, while attraction increases as the gap shrinks.

Formula/Principle: Use force balance for equilibrium and the sign of the net-force derivative for stability.

Step 1: Force balance gives

$$V^2 = \frac{2k}{\epsilon_0 A}x(d - x)^2.$$

Step 2: For stability the derivative of net inward force must be negative:

$$\frac{\epsilon_0 A V^2}{(d - x)^3} - k < 0.$$

Step 3: Substitute the equilibrium relation to obtain

$$\frac{2kx}{d - x} < k \Rightarrow 3x < d.$$

Step 4: At x=d/3, the equilibrium voltage reaches its maximum.

Worked derivation:

$$V_{lim}^2 = \frac{2k}{\epsilon_0 A}\frac{d}{3}\left(\frac{2d}{3}\right)^2 = \frac{8kd^3}{27\epsilon_0 A}.$$

Why: Force balance only locates stationary positions; stability additionally tests whether a small inward perturbation is restored or amplified. Equality is marginal, so it is not included in the strict stable interval.

Why not the others: (A) ignores destabilising gap feedback. (B) retains the unstable branch. (D) uses the wrong displacement threshold and therefore wrong voltage.

Key point: Test the force slope after solving equilibrium in a voltage-controlled electromechanical system.

Q54 / A Susceptibility sign, thermal response and remanence distinguish dia-, para- and ferromagnetism

Answer: (A) The sign, temperature trend and memory distinguish the three magnetic responses.

Given: In the linear weak-field description, relative permeability is 1+chi.

Formula/Principle: Classify response using susceptibility sign, temperature dependence and remanence; relative permeability is 1+chi.

Step 1: Negative weak susceptibility identifies P as diamagnetic; positive inverse-temperature behaviour identifies Q as paramagnetic.

Step 2: Strong response with remanence below a transition is consistent with ferromagnetism in R.

Worked derivation: For P,

$$\mu_r = 1 + \chi < 1 \quad (\chi < 0).$$

Why: Remanence supplies evidence beyond the mere sign of susceptibility, while a small negative response is not equivalent to weak paramagnetism. Heating beyond the ferromagnetic transition can remove long-range ordering without making the material diamagnetic by definition.

Why not the others: (B) exchanges signs and confuses magnetic and carrier classifications. (C) contradicts the stated weak P response and positive Q response. (D) discards the decisive sign difference.

Key point: Combine susceptibility sign, thermal dependence and hysteretic memory rather than magnitude alone.

Q55 / A Even nonuniform line charge cancels horizontal field while vertical components reinforce

Answer: (A) The field is perpendicular to the rod toward positive y.

Given: Charge density is even in x, and P is above the rod.

Formula/Principle: Pair equal source elements by reflection, then add each vector component with its sign.

Step 1: Elements at x and $-$x have equal positive charge and equal distance to P. Their x components are opposite.

Step 2: Both y components point upward because each positive element repels a positive test charge away from the rod.

Worked derivation:

$$E_x = 0, \quad E_y = 2k \int_0^L \frac{\lambda_0 x}{L} \frac{a\,dx}{(x^2 + a^2)^{3/2}} > 0.$$

Why: Symmetry removes only the component that changes sign under reflection. Vanishing local charge density at the midpoint does not erase the contribution of the rest of the rod. No field-magnitude evaluation is needed for the direction asked.

Why not the others: (D) ignores equal density growth toward the left end. (B) reverses the direction from positive sources. (C) assumes every vector component cancels.

Key point: Apply symmetry component by component, especially for nonuniform but even charge distributions.

Q56 / C Two equal-load-power branches have different internal losses and source-energy costs

Answer: (C) The higher-resistance branch minimises source energy, which is about 45.84 J.

Given: Required load power is 40/10=4 W.

Formula/Principle: Impose the delivered-power constraint first, then compare emf times current times time on its two resistance branches.

Step 1:

$$4 = \frac{6^2 R}{(R+1)^2} \Rightarrow R^2 - 7R + 1 = 0,$$

giving R=(7±3sqrt(5))/2 ohm.

Step 2: Source energy is emf times transported charge:

$$E_{src} = 6It = \frac{360}{R+1}.$$

It decreases as R increases, so choose the larger root.

Step 3: Substitute it:

$$E_{src} = \frac{720}{9+3\sqrt{5}} = 180 - 60\sqrt{5}\ \text{J}.$$

Worked derivation:

$$E_{internal} = E_{src} - 40 \approx 5.84\ \text{J} > 0.$$

Why: Equal delivered load energy and time fix the output power but do not fix current. The high-resistance branch carries less current and wastes less energy in the internal resistor. Maximum-power matching is not the task here.

Why not the others: (A) chooses the high-loss branch and omits internal heating. (B) maximises load power rather than meeting the specified 4 W. (D) equates delivered and source energies despite nonzero internal resistance.

Key point: Optimise the stated energy cost under the delivery constraint, not an unrelated maximum-power criterion.

Q57 / B A finite eye–lens separation changes the near-point image geometry of a simple magnifier

Answer: (B) The object is 4.0 cm in front of the lens and angular magnification is 5.

Given: The virtual image is 25 cm in front of the eye but only 20 cm in front of the lens, so v=−20 cm.

Formula/Principle: Use the image distance from the lens in the lens equation and the image distance from the eye in the angular comparison.

Step 1: With the Cartesian lens formula,

$$1/5 = 1/(-20) - 1/u \Rightarrow u = -4\ \text{cm}.$$

Step 2: The virtual image magnification is

$$m = v/u = (-20)/(-4) = 5.$$

Step 3: Image–eye distance and unaided comparison distance are both 25 cm, so their angular ratio equals the image/object height ratio.

Worked derivation:

$$M = (5h/25)/(h/25) = 5.$$

Why: The standard near-point expression 1+D/f assumes the eye is effectively at the lens. Its image-distance input must be corrected when the eye is separated by a stated distance.

Why not the others: (A) ignores eye–lens separation. (C) finds the object distance correctly but imports the zero-separation magnification. (D) places the object at focus, giving an image at infinity instead of the near point.

Key point: Keep distances measured from the eye separate from those measured from the lens.

Q58 / D Contact voltage shifts a stopping-potential reading without changing the spectral emission threshold

Answer: (D) The reported reading is 1.50 V, and only the 300 nm light exceeds threshold.

Given: Contact offset changes the electrical reading, not the energy required to free an electron.

Formula/Principle: Maximum kinetic energy is photon energy minus work function; the electrical offset changes only the readout.

Step 1: At 400 nm,

$$E_\gamma = 1240/400 = 3.10 \text{ eV}, \quad K_{max} = 3.10 - 2.00 = 1.10 \text{ eV}.$$

Step 2: True stopping potential is 1.10 V, so the reading is 1.10+0.40=1.50 V.

Step 3: Threshold wavelength is

$$\lambda_{th} = 1240/2.0 = 620 \text{ nm}.$$

Hence 800 nm fails and 300 nm succeeds.

Worked derivation:

$$E_{800} = 1.55 \text{ eV} < 2.0, \quad E_{300} \approx 4.13 \text{ eV} > 2.0.$$

Why: A measurement offset can move a fitted voltage intercept without changing the physical photon threshold. Increasing electrical readout alone cannot supply missing incident photon energy in this model.

Why not the others: (A) omits the offset and permits subthreshold infrared. (B) applies the voltage result correctly but ignores the wavelength threshold. (C) subtracts the offset and reverses the spectral ordering.

Key point: Separate an instrumental voltage offset from the material's energy threshold.

Q59 / B An initially excited atom can emit more than its latest absorbed photon without violating energy conservation

Answer: (B) The return emits 12.09 eV in total, drawing on both the new photon and the initial excitation.

Given: The atom begins above the ground state before the stated absorption.

Formula/Principle: Total emitted photon energy equals the drop from the atom's complete initial internal energy to the final state.

Step 1: Energy after absorption relative to ground is

$$10.20 + 1.89 = 12.09 \text{ eV}.$$

Step 2: Returning to n=1 releases that entire difference, whether directly or through n=2.

Worked derivation: A two-step cascade emits

$$E_{32} + E_{21} = 1.89 + 10.20 = 12.09 \text{ eV}.$$

Why: Energy accounting must include the initial atomic state, not only the latest incident photon. The total emitted energy does not depend on how the permitted gaps are partitioned into photons. The atom cannot repeat the same absorption starting from n=1 without being prepared again.

Why not the others: (A) ignores stored excitation. (C) ignores the newly absorbed energy. (D) counts the full gap twice merely because two photons can be emitted.

Key point: Include the initial internal energy when comparing absorbed and emitted radiation.

Q60 / C AC generator polarity reverses while resistive-load torque continues to oppose rotation

Answer: (C) Current reversal does not reverse the direction of energy conversion in a resistive generator.

Given: If flux linkage is K cos theta, emf is K omega sin theta and current is that emf divided by R.

Formula/Principle: Use Faraday emf, resistive current and the current–geometry torque product; drive work supplies Joule heat.

Step 1: Magnetic torque on the loop contains the product of current and sin theta.

Step 2: Its resisting magnitude is

$$|\tau| = \frac{K^2\omega}{R}\sin^2\theta.$$

Worked derivation:

$$P_{drive} = |\tau|\omega = \frac{K^2\omega^2}{R}\sin^2\theta = I^2R \geq 0.$$

Why: Both current and the relevant geometrical factor change sign after half a turn, leaving the resisting torque direction unchanged. A negative instantaneous emf specifies polarity, not negative dissipated power. At the stated nonzero instants the drive must supply positive work.

Why not the others: (A) ignores the simultaneous geometrical sign reversal. (B) treats a signed current as absence of current. (D) contradicts the positive Joule-energy demand of an ideal resistive load.

Key point: Infer torque and power from the full sign product, not from emf polarity alone.